
Super Structured Quad Mesh

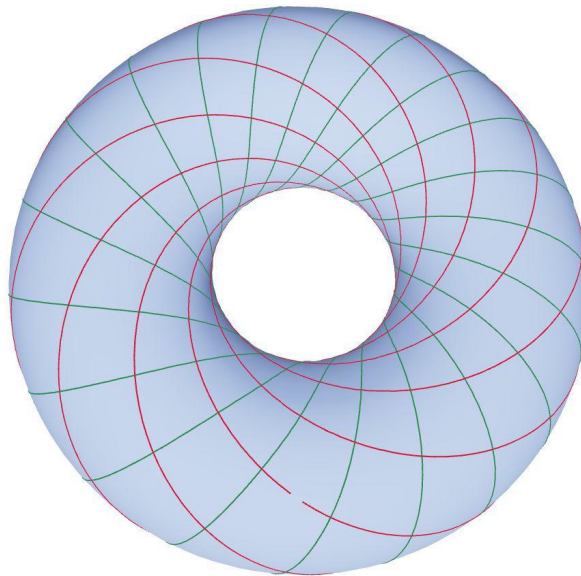
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26 / 08 / 2025

Abstract: This paper proposes a brand-new concept and research direction of super-structured quadrilateral meshes. A super-structured quadrilateral refers to a quadrilateral arrangement whose global quad structure is designable and controllable. Outline the differences and innovations between the concept of super-structured quadrilateral meshes and that of ordinary structured quadrilateral meshes. The construction of super-structured quadrilateral meshes relies heavily on overall geometric structures, especially the geometric structures and concepts on 2D surfaces. Specify the general scope of key differential geometry and topology theories required for the implementation of super-structured quadrilateral meshes, such as moduli spaces, Teichmüller spaces, and sheaves. Present some preliminary algorithms and visualized experimental results from the research on super-structured quadrilateral meshes. Discuss the necessity of super-structured quadrilateral meshes in industrial software, such as their potential to solve technical challenges in hexahedral mesh generation on high-genus meshes, isogeometric analysis, spline surface generation, T-splines, and finite element computation. Present some of the authors' relevant work on super-structured quadrilateral meshes as well as computational discrete global geometric structures. This paper also proposes that super-structured quadrilateral meshes can serve as an example demonstrating how cutting-edge geometry and topology theories play a critical role in industrial technology; conversely, the industrial application of these meshes can provide impetus for the further research and development of related geometry and topology theories. Inspired by the "Holy Grail Problem" in hexahedral meshing, this paper puts forward the "Mountain Lingshan Problem" for quadrilateral meshing.

Key Words : Quadrilateral mesh generation, Finite element computation, Arrangement structure, Global geometric structure, Harmonic foliation structure, Teichmüller space, Moduli space, Square tiling on surfaces, Dynamical system

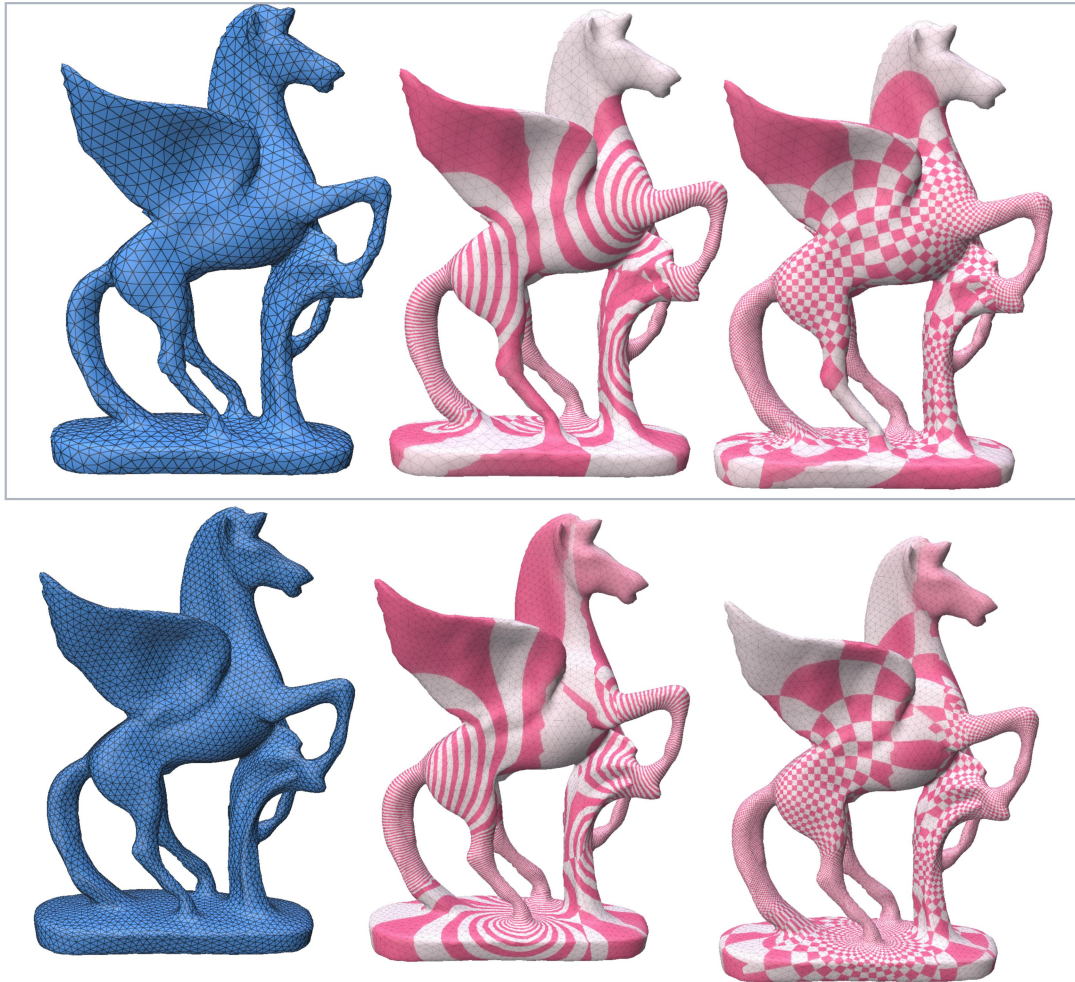


(Figure 1: Super structured quad mesh, generated by Hui Zhao with the software Geometric .)

1. Concepts and Definitions

Building on previous research, the author of this paper proposed the novel concept of "Super Structured Quad Mesh" featuring "controllable and designable global arrangement of quadrilaterals" in 2023, and defined its research directions, research fields, research content, research tools, and the applied geometric topology theories.

In both academic and industrial circles, the Structured Quad Mesh is relatively well-known. Mesh generation is generally categorized into three types: unstructured meshes (as shown in the left figure of Figure 2), structured meshes (as shown in the right figure of Figure 2), and hybrid meshes. Specifically, an unstructured mesh refers to a mesh whose surface is composed of triangular elements and interior is filled with tetrahedral elements; a structured mesh refers to a mesh whose surface and interior are completely filled with quadrilateral elements and hexahedral elements, respectively; and a hybrid mesh is a mesh composed of a mixture of surface elements (such as triangular, quadrilateral, and pentagonal elements) and volume elements (such as tetrahedral and hexahedral elements).

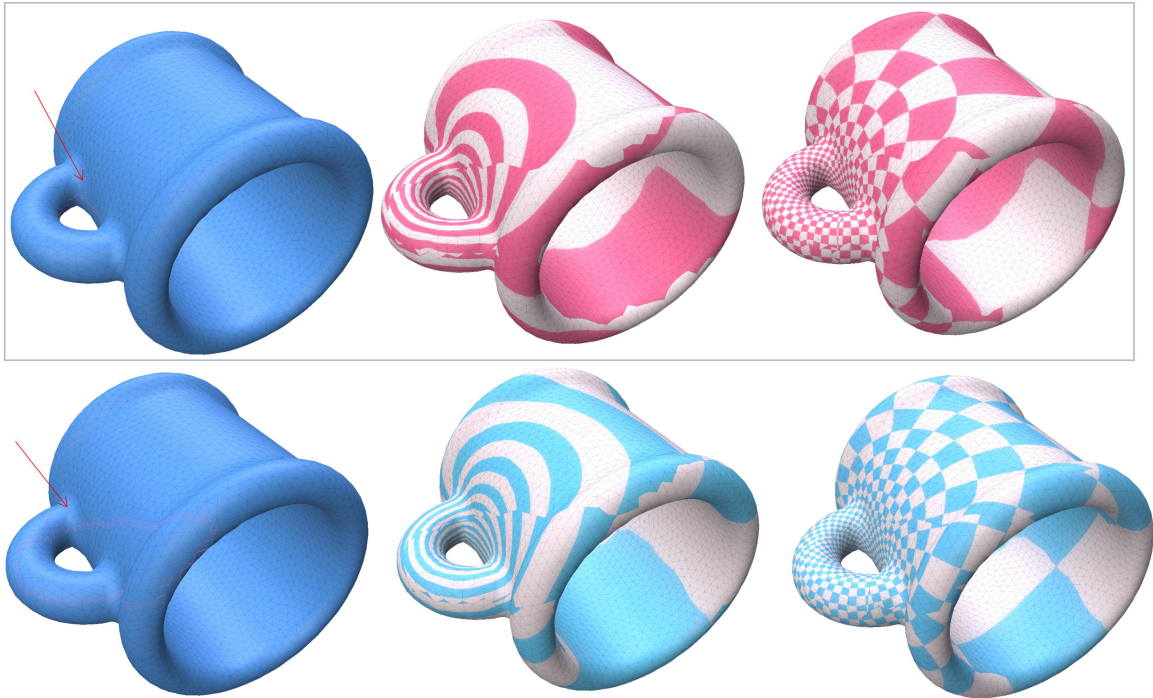


(Figure 2: left is triangle mesh, right is structured quad mesh, generated by Geometric .)

Structured quad mesh generation can be further subdivided into four categories: Regular, Semi-Regular, Valence Semi-Regular, and Irregular. However, this subdivision method lacks clear and

rigorous criteria, and is usually roughly defined based on the number of blocks obtained from the meshing. Generally speaking, the fewer the number of blocks, the higher the regularity of the mesh. This means that there is currently no algorithm available to quantitatively determine the regularity degree (regular or irregular) of a given structured mesh; in practice, the optimization goal is mostly "the fewer the number of blocks, the better". To make a strict judgment on regularity, it is necessary to study the meshing method, which is exactly the research content of the super structured quad mesh proposed in this paper—an area that has not yet been explored by the academic and industrial communities.

A single 3D model can correspond to countless meshing schemes, as shown in the meshing results in the first and second rows of Figure 3. Even for structured quad meshes, a single model can still have numerous structured meshing results, with each result featuring a distinct overall arrangement structure of quadrilaterals. The arrangement can be the common one shown in Figure 4, or the circular arrangement presented in Figure 1.



(Figure 3: Left is triangle mesh, right is structured quad mesh, generated by geometric .)

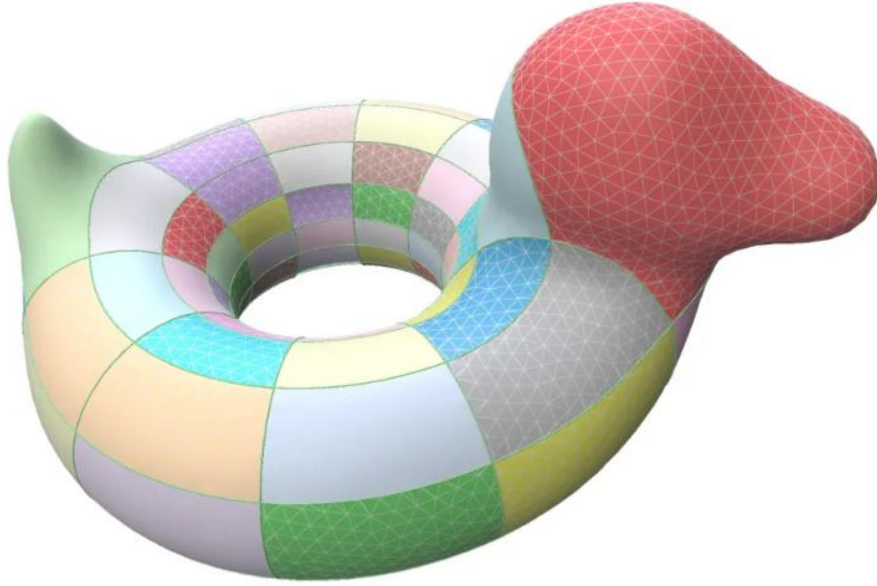
However, various quad mesh generation algorithms in current academic research can only randomly produce a certain meshing result, and are unable to precisely control or pre-design specific arrangement structures. The research field of super structured quad mesh proposed in this paper focuses precisely on how to design and control the generation of specific overall arrangement structures. The core connotation of the prefix "super-" in this concept refers to the controllability and designability of the overall arrangement structure of quadrilaterals.

This paper clearly points out that the core of a quad mesh does not lie in the fact that each local element is a quadrilateral, but in the overall structure formed by the combination of these quadrilaterals. This is an aspect that has not been paid attention to or clearly described in the current academic community, and thus there is no mention of in-depth research and solutions for it.

For example, a triangular mesh can be converted into a quad mesh by removing the adjacent edges of two triangles and merging them into a single quadrilateral. However, such a quad mesh has no overall

regularity at all, and therefore is not fundamentally different from the original triangular mesh. Hence, when studying the generation of quad meshes, this paper proposes that the focus should be on the generation of the overall arrangement structure of quad meshes.

The overall arrangement structure of a quad mesh contains the regional division of the entire mesh. In current academic research, part of the structural thinking for the structured quad meshes under discussion starts from the perspective of block division. However, there is still no rigorous definition for clear block division—specifically, there is no clear definition of what constitutes good block division or poor block division. As a result, there is no clear algorithm to obtain an optimized block division method; at present, the quality of block division is only defined based on the criterion of the number of blocks.



(Figure 4: Super structured quad mesh, generated by Geometric .)

This paper points out that the regional division of quad meshes must be based on the research of the overall arrangement structure of quadrilaterals, i.e., the research on super structured quad meshes. Only when the research on super structured quad meshes is realized can other problems related to quad meshes be solved smoothly.

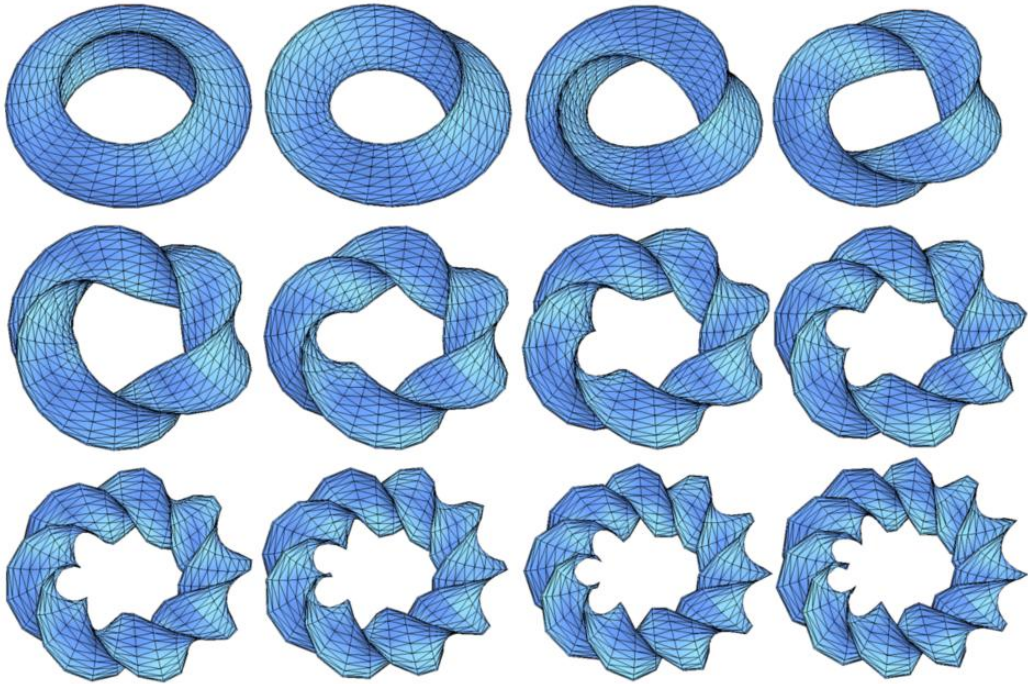
For triangular meshes, there is no constraint on such an overall arrangement; it is sufficient as long as the local quality of the generated triangles is good. However, the particularity of quad meshes lies in the "overall constraint," which constitutes an integral geometric structure. Currently, a large part of the ideas, concepts, and terminology in academic research on quad mesh generation is derived from the research on triangular mesh generation — specifically, the pursuit of the quality of triangles and quadrilaterals. This paper proposes that, on the premise of achieving controllability over the overall arrangement of quadrilaterals, efforts should then be made to address technical issues related to local elements, such as the quality of individual quadrilaterals and feature lines.

The research on super structured quad meshes differs significantly from the current research on quad meshes in terms of objectives, concepts, ideas, methods, and applied geometric topology theories. It requires investigation from entirely new logics, approaches, and perspectives.

The research motivation for super structured quad meshes stems from two aspects: first, the algorithm design for various overall geometric structures in differential geometry and topology theories; second, the demand-driven need to solve technical problems faced by current CAD/CAE industrial software that are difficult to address with existing algorithms.

Much of the algorithmic research on triangular meshes relies on classical differential geometry developed before the 1950s, whereas the research on quad meshes needs to be based on the mathematical theory of global geometry pioneered by Shiing-Shen Chern after the 1950s. Although the shift from triangular meshes to quad meshes may seem like a small step on the surface, it represents a huge leap in terms of the underlying mathematical theories—this is the subtlety of scientific research exploration.

The research on super structured quad meshes is not yet concluded; instead, it is still in its initial stage. It is not a single algorithm but a complete research direction and field. The realization of super structured quad meshes requires algorithmic research on a large number of differential geometry and topology theories, some of which are still under investigation in the mathematical community.



(Figure 5: Umbilic Torus.)

Mountain Lingshan Problem

There is a well-known "holy grail" problem in the field of mesh generation, which refers to the question of how to automatically generate high-quality structured hexahedral meshes for a given 3D spatial domain with complex topology and geometry. Hexahedral meshes have advantages in simulating physical phenomena involving large deformations, but their generation algorithms and theories are still immature, requiring manual intervention—this problem is known as the "holy grail" challenge in the mesh research community. The "holy grail" problem in meshing does not have a clear single proposer; instead, it gradually emerged as a widely concerned and important issue during the development of the mesh generation field. (The "holy grail" is a hexahedral mesh, i.e., a decomposition of the domain into conforming cube-like elements.)

This paper defines a "Mountain Lingshan" problem: For 2D surfaces of arbitrary genus, how to generate quadrilateral meshes with controllable and designable overall quadrilateral arrangement structures?

The "holy grail" problem focuses on 3D hexahedral meshes, while the "Mountain Lingshan" problem focuses on 2D quadrilateral meshes.

The Laboratory of computational discrete Global Geometric Structures proposed the "Mountain Lingshan" problem based on the research direction of "super structured quad meshes," and this research direction provides a solution pathway for the "Mountain Lingshan" problem. Currently, there is no roadmap for solving the "holy grail" problem.

The solution to the "Mountain Lingshan" problem will contribute to solving the "holy grail" problem. This is because the surface of a hexahedral mesh consists of quadrilaterals; if the overall arrangement structure of the surface quadrilateral mesh can be designed, the way to divide the internal hexahedral mesh will become clearer.

The relationship between the "Mountain Lingshan" problem and the "holy grail" problem is similar to that between the Poincaré Conjecture and the Thurston Conjecture—once the Thurston Conjecture was solved, the Poincaré Conjecture was automatically proven.

The naming of the "holy grail" is derived from Western Christian culture, while the naming of "Mountain Lingshan" comes from traditional Chinese culture. The naming of the "holy grail" problem has inspired and united research on hexahedral mesh generation over the past 30 years. Similarly, the naming of the "Mountain Lingshan" problem will inspire and unite research on quadrilateral mesh

generation in the next 30 years.

The name "Lingshan" originates from the highest peak in Beijing, with an altitude of 2,303 meters. The Lingshan Natural Scenic Area is surrounded by mountains, with crisscrossing terrain and gullies. Its northern main peak reaches an altitude of 2,303 meters, making it the highest peak in Beijing and forming a geographical commanding height of the city. Influenced by the altitude, within an area of approximately 25 square kilometers, Lingshan exhibits the unique natural landforms of fault-block mountains and folded-mountain peaks in the Beijing region. The main topographic features of the Lingshan Scenic Area are folded mountains and fault-block mountains.

For the simplest 2D case of the Umbilic Torus, the "Mountain Lingshan" problem has a formula-based solution, as shown in the figure below, which is very intuitive.

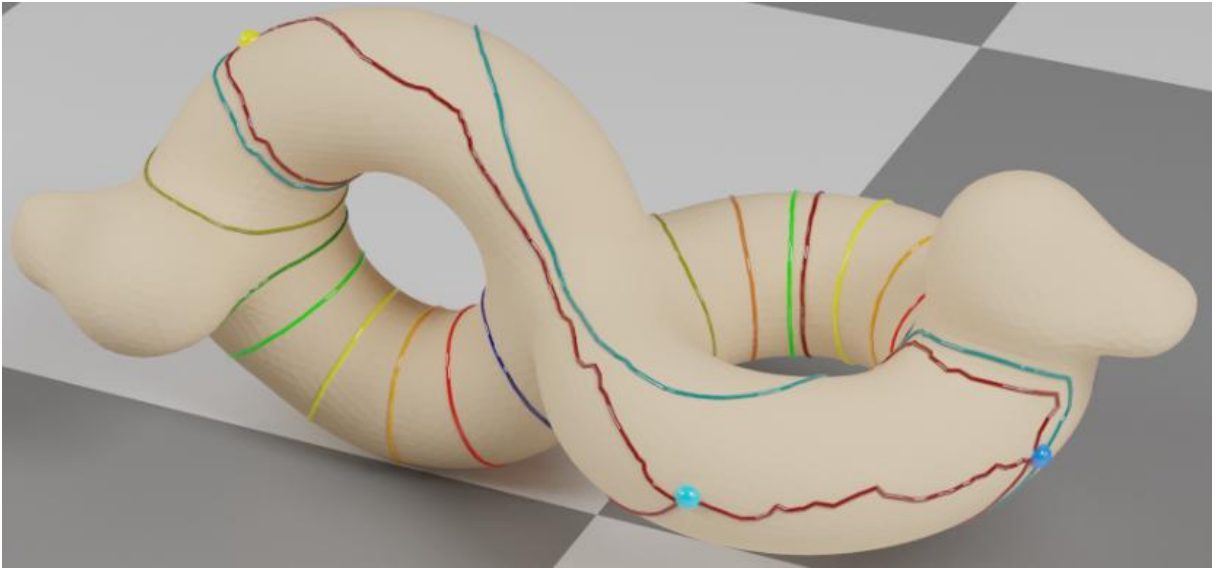
2. The required differential geometry and topology theories

Research on unstructured triangular mesh generation has a history of decades both domestically and internationally, resulting in a large number of classical algorithms. In recent years, many efficient algorithms have also emerged in the field of structured quadrilateral mesh generation, which are based on theories such as vector fields, frame fields, global seamless parameterization, flat metrics, Ricci flow, and Calabi flow.

In the research process from unstructured triangular meshes to structured quadrilateral mesh generation, the geometric and topological concepts and theories applied by the academic community have been continuously deepened and expanded. Starting from basic high school geometry knowledge, it has gradually extended to classical differential geometry and even cutting-edge achievements in contemporary differential geometry, showing a trend of continuous increase in theoretical complexity.

The development of Euclidean plane geometry has spanned thousands of years. After the emergence of calculus, local differential geometry (Differential Geometry)—which applies calculus to the study of curves and surfaces in three-dimensional space—came into being.

The founder of global geometry (Global Geometry) is the mathematics master Shiing-Shen Chern. Around the 1930s, before Mr. Chern's work, the research on local differential geometry was already relatively sufficient, and many scholars at that time believed that there was very limited research space left in this field. It was not until the 1940s that Shiing-Shen Chern, a master of mathematics, pioneered the brand-new field of "global geometry" through groundbreaking work such as the proof of the Gauss-Bonnet theorem and the construction of Chern classes. Its core lies in establishing a connection between local geometric information and global topological properties, which can be regarded as a breakthrough development in traditional local classical differential geometry (Differential Geometry). Currently, research related to global geometry is still developing rapidly; for example, mathematics masters such as Shing-Tung Yau and Gang Tian have made many important breakthroughs in aspects such as the existence of global geometric structures using tools of geometric analysis.



(Figure 5: A harmonic foliation and Ribbon graph, generated by Geometric .)

One primary reason why the concept of super structured quad meshes was not proposed until recent years is that achieving the controllability and designability of the overall quadrilateral arrangement structure relies on a series of new geometric and topological theories continuously explored by mathematicians over the past two decades — especially the work on 2D surfaces by mathematicians represented by Fields Medal laureate William Thurston, as shown in the figure.

This paper only outlines the general scope of geometric and topological theories and mathematicians involved in the realization of super structured quad mesh research, rather than providing a detailed list. It merely serves as a preliminary index, functioning as an introductory guide to facilitate in-depth learning.

- (1) Moduli space
- (2) Teichmüller space
- (3) Dynamical systems on surfaces
- (4) Ribbon-Graph
- (5) Harmonic foliations
- (6) Holomorphic quadratic differentials
- (7) Meromorphic quadratic differentials
- (8) Thurston norm
- (9) Translate surface
- (10) Half translation surface
- (11) Flat surface
- (12) Riemann surface
- (13) Square-tiled surfaces
- (14) Masur-Veech volume
- (15) Meanders
- (16) Billiards
- (17) Interval exchange
- (18) Teichmüller flow
- (19) Strata of abelian and quadratic differentials
- (20) Square-Tiling on surfaces
- (21) Mapping class groups of surfaces

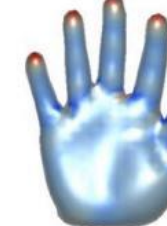
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Many of the aforementioned geometric structures belong to global geometric structures, which are closely associated with topology and defined globally on surfaces. In contrast, local geometric structures such as Gaussian curvature and mean curvature are only defined locally.

The logical concept of "controllable and designable overall quadrilateral arrangement structure" proposed in this paper needs to be translated into the language of the aforementioned differential geometry and topology. To elaborate on and explain the controllability, designability, and overall arrangement patterns of quadrilaterals in detail, it is necessary to connect the theorems and concepts of differential geometry and topology mentioned above with quadrilateral arrangement patterns. Notably, "controllable and designable overall quadrilateral arrangement structure" does not correspond to a single theorem; instead, it requires the integration of many of the aforementioned theorems and concepts for description, making it a large-scale systematic project. In terms of workload estimation, it would approximately require research efforts equivalent to 100 academic papers.

The research direction of super structured quad meshes aims to achieve this goal, converting practical application requirements into rigorous mathematical language. One of the contributions of this paper is

explicitly proposing this scientific problem, identifying the required geometric and topological theories, presenting preliminary work, and outlining future research approaches.

原始模型	Gaussian 曲率	平均曲率	最大主曲率	最小主曲率
				
				
				
				

(Figure 6: Several kinds of curvatures, curvature numerical value are colored.)

The research on super structured quad meshes focuses on 2D meshes, which correspond to 2D surfaces. Therefore, the key focus should be on geometers whose work centers on 2D surfaces, including the following scholars (several of whom are Fields Medal laureates):

- (1) Bill Thurston
- (2) Curtis McMullen
- (3) Maxim Kontsevich
- (4) Maryam Mirzakhani
- (5) Anton Zorich
- (6) Alex Eskin
- (7) Scott A. Wolpert

-
- (8) Howard Masur
 - (9) Benson Farb
 - (10) Alex Wright
 - (11) Vincent Delecroix
 - (12) Jayadev S. Athreya
 - (13) Carlos Matheus
 - (14)

In China, there are also scholars engaged in research on differential geometry and topology in this field. For instance, Professor Yu Fei from Zhejiang University and Professor David Chen from Boston University organized the [2024 Academic Symposium on Geometry and Moduli Spaces](#).

The innovation of the super structured quad meshes proposed in this paper lies in connecting the mathematical fields studied by these geometers with subsequent fields such as mesh generation, finite element computation, and mechanical analysis. The number of scholars specializing in geometric and topological theory research is relatively small, whereas the industrial software field that uses meshes encompasses millions of engineers, practitioners, and enormous technical demands. Through the research on "algorithms" for super structured quad meshes, experts from these two sectors (theoretical and industrial) can be integrated into a team with shared goals. This not only enables the development of an interdisciplinary intersection of differential geometry and topology, computer science, mechanical applications, and industrial software but also, in turn, promotes research on relevant abstract mathematics through specific technical applications and attracts more people to engage in work in this field.

几何与模空间

2024年7月1-12日@浙江大学

会议动机: 几何学在数学中占据着至关重要的地位,通过运用代数、分析、拓扑等多种方法研究几何,实现了将数学的各个领域紧密地联系在一起。模空间作为几何研究的核心之一,扮演着连接相似几何结构的纽带角色。近年来,几何与模空间的研究取得了许多关键性突破,同时也催生了一系列新的研究方向。在这一背景下,我们计划于二零二四年七月在浙江大学举办为期两周的“几何与模空间”学术会议。旨在介绍相关领域的前沿成果,促进来自不同学术背景的学者之间的交流,并致力于培养年轻学者的研究成长。我们期望,此次会议将成为一个有益的合作平台,为数学领域的新发展提供契机。

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On the surface, achieving the controllability and designability of the overall quadrilateral arrangement structure seems to be merely an optimization of the random generation results of structured quadrilateral meshes; in reality, however, it requires a large number of brand-new geometric and topological theories as support to complete the algorithm design. This demands the construction of discrete algorithms for the aforementioned geometric and topological theories, as well as the realization of robust computation for

their corresponding specific values.

The research on super structured quadrilateral meshes needs to take the theory of square-tiling on surfaces as the core and integrate the various aforementioned theories to explore algorithmic research. These theories are intertwined and interdependent; therefore, during the research on super structured quadrilateral meshes, it is necessary to solve a great many research tasks step by step—including those related to relevant theories, concepts, algorithms for geometric structures, and visualization—rather than directly achieving quadrilateral meshes with controllable and designable overall arrangement structures in one fell swoop.

3. Graphic Presentation of Existing Work on Super - structured Quadrilateral Meshes

In the "super structured quad mesh" proposed by the author, the concept of "controllable and designable overall arrangement structure of quadrilaterals" is an entirely new idea, rather than an existing research achievement in the mathematical community. Therefore, its realization requires the comprehensive application of various existing "global geometric structures" and other related geometric structures in the field of geometric topology.

Specifically, the construction of the "super structured quad mesh" does not rely on the discretization of a single global geometric structure; instead, it necessitates the combination of discretization algorithms for a large number of "global geometric structures." This research involves conducting theoretical exploration and algorithm design for the various geometric structures mentioned earlier, followed by code implementation, and further exploring the technical pathway to achieve the goal of "controllability and designability." Currently, this research is still in the exploratory stage.

In the research direction and field of super structured quad meshes, although the ultimate goal is to realize the Square Tiling (square tiling) of surfaces, Square Tiling itself depends on other theories and structures. We are designing algorithms for each geometric topology concept one by one. Among these, the most advanced progress in algorithm design has been made for harmonic foliations.

A quad mesh can be regarded as consisting of a set of horizontal harmonic foliations and a set of vertical harmonic foliations; thus, the algorithmic research on harmonic foliations serves as the foundation. Moreover, the algorithmic exploration of harmonic foliations involves the computation of structures such as Ribbon-Graph. It is evident that the computation of a specific geometric structure requires the prior research of other related structures, which makes the dozens of aforementioned geometric topology concepts and theories form an interconnected and organic whole.

The algorithmic workflow for harmonic foliations is as follows: starting from the input non-harmonic foliations, a Ribbon-Graph is obtained through constrained optimization, and then further optimization is performed to achieve the harmonic result; holomorphic quadratic differentials, on the other hand, can be obtained by rotating the input horizontal harmonic foliations by 90 degrees on the dual model.

This paper only provides a brief overview of the relevant intuitive descriptions and algorithmic experimental results for the involved mathematical concepts and theories; for more detailed and specific mathematical concepts, please refer to relevant textbooks.

The research on super structured quad meshes requires algorithm design for the aforementioned large number of differential geometry and topology theories. This paper presents algorithm diagrams of some geometric concepts for which preliminary research results and progress have been achieved:

- (1) Non-harmonic foliations
- (2) Harmonic foliations
- (3) Ribbon-Graph
- (4) Optimization from non-harmonic to harmonic
- (5) Foliations with poles
- (6) Holomorphic quadratic differentials
- (7) Meromorphic quadratic differentials

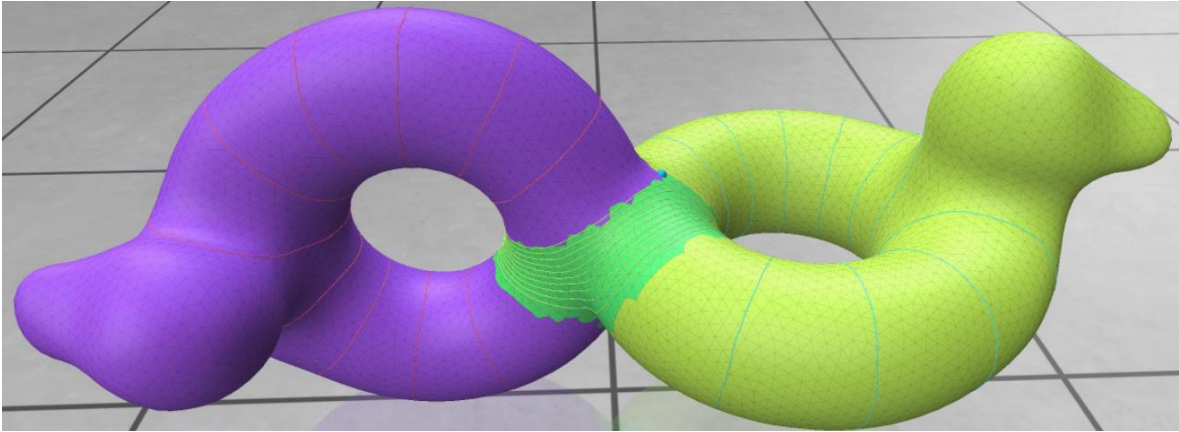
(8) Harmonic foliations on surfaces with boundaries

(9) Super structured quad meshes

Through the algorithm design, code implementation, and visual observation of these global geometric structures, as well as the comparison with differential geometry and topology theories, we can gradually understand the nature, characteristics, technical tools, and theoretical scope of the research direction of super structured quad meshes.

(1) Non-Harmonic Foliation

Foliations can be either harmonic or non-harmonic. A harmonic foliation refers to a set of the smoothest parallel lines under the current metric, while a non-harmonic foliation consists of non-smooth lines. As shown in the figure below, the lines of the non-harmonic foliation are relatively dense in the middle of the mesh model and sparse in other areas; in contrast, the lines of a harmonic foliation would be evenly distributed across the surface. At each local point on the surface, a non-harmonic foliation cannot be represented by a harmonic function—that is, it does not satisfy the conditions of a harmonic function, hence the name "non-harmonic foliation". However, it is topologically equivalent to a certain harmonic foliation.



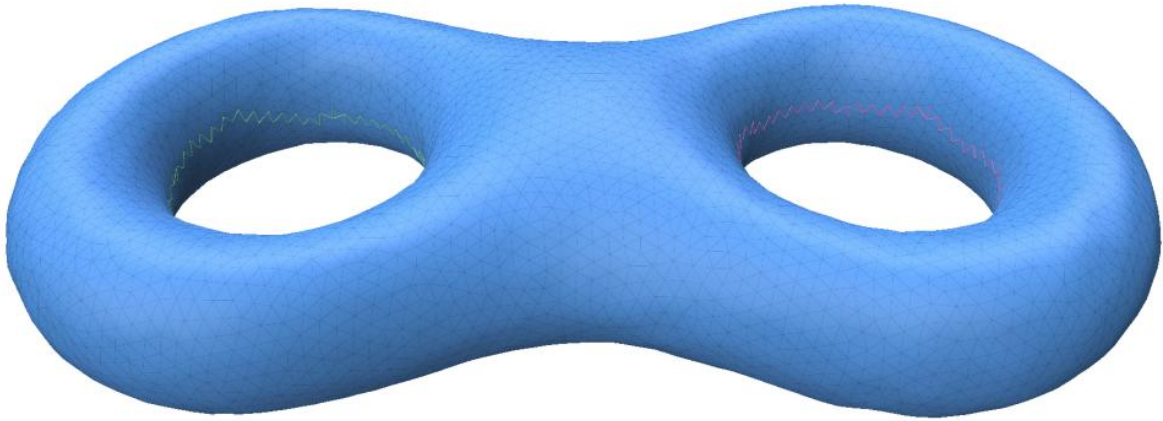
(Figure 7: A non-harmonic foliation, generated by Geometric.)

A non-harmonic foliation may also fail to cover the entire surface and only exist in a certain part of the surface. As shown in the figure, the two annuli of different colors represent a foliation—this foliation is not only non-harmonic, but also, in the regions outside the annuli, the local values of the foliation are all zero, with non-zero values only present on the annuli.

However, through optimization, the regions with non-zero values can be gradually expanded to the regions with zero values. Nevertheless, these foliations are all equivalent in the sense of Whitehead Classes.

This phenomenon reflects the characteristic of the global geometric structure of foliations: even though the values in most regions are zero, the foliation in the initial region has already determined the topology of the entire foliation.

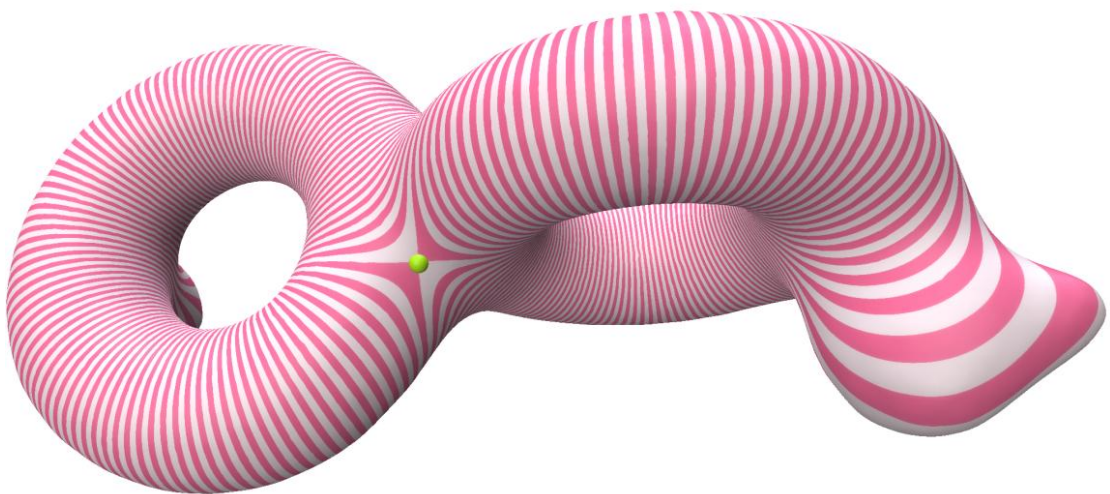
This characteristic cannot be studied from other perspectives such as discrete exterior calculus. Therefore, this is why the author proposes that "algorithm design" should be considered from the perspective of "global geometric structures"—thereby enabling the development of more algorithms and applications related to global geometric structures.



(Figure 8: A non-harmonic foliation, generated by Geometric.)

(2) Harmonic Foliations

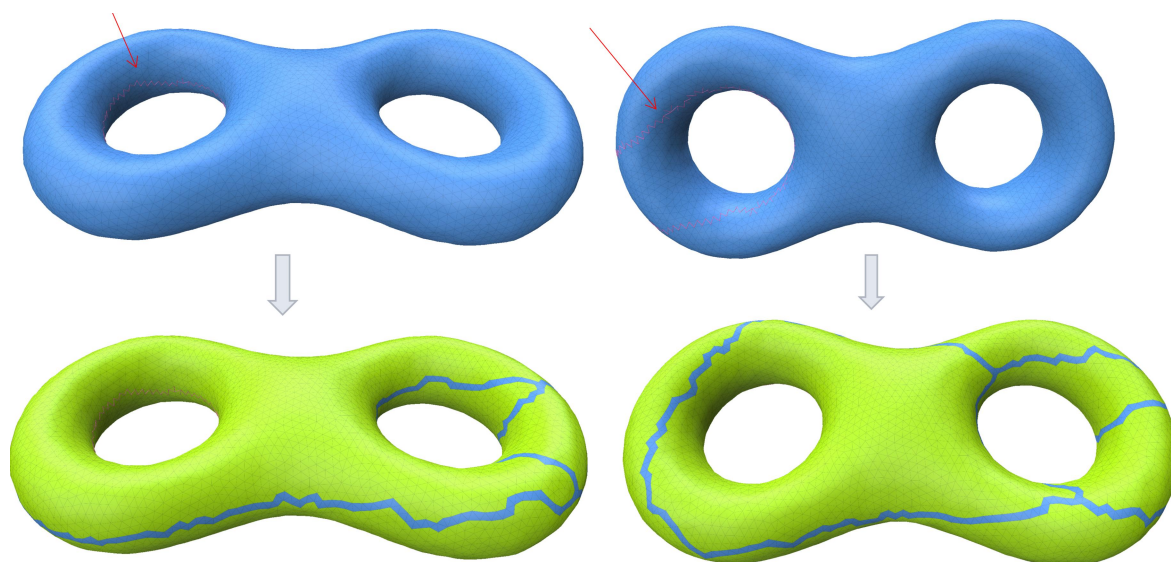
Every point of a harmonic foliation can be represented by a local harmonic function, but this requires multiple harmonic functions expressed in multiple coordinate systems as well as the transformations between them. A harmonic foliation cannot be represented by a single harmonic function on the entire surface—this is the global property of a global geometric structure. As shown in the figure, the lines on the harmonic foliation are evenly distributed, achieving the smoothest result under the metric of the surface. If the genus of the surface is not 1, zeros (i.e., points where the values of several independent local functions are zero) will appear when the foliation covers the entire surface. The zero value of these zeros does not change with the coordinate system and remains zero in any coordinate system, making them special points. However, the number of zeros of a foliation on each surface is finite and conforms to a specific formula. The small spheres in the figure represent the zeros of the harmonic foliation. On a surface with genus g , a harmonic foliation has $(4g - 4)$ zeros of degree 1. Three lines meet at a zero of degree 1. Two zeros of degree 1 can merge into a zero of degree 2, at which four lines meet—just like the zero shown in this figure. For a non-harmonic foliation that covers the entire surface, there are also $(4g - 4)$ zeros.



(Figure 9: A harmonic foliation, generated by Geometric.)

(3) Ribbon-Graph

A Ribbon-Graph is a special type of graph on a surface. Each annulus of a non-harmonic foliation can generate several Ribbon-Graphs, as shown in the figure: the red arrows indicate the foliation, while the white arrows indicate the generated Ribbon-Graphs. Ribbon-Graphs have many important applications in the research of various geometric theories; here, they are used in the optimization process of foliations from non-harmonic to harmonic.



(Figure 10: A foliation and a ribbon graph, generated by Geometric.)

(4) The Optimization of Non-Harmonic to Harmonic Foliation

A non-harmonic foliation can be optimized into a harmonic foliation via an "algorithm". As shown in the figure below, the non-harmonic foliation in the upper left (a blue annulus) is gradually optimized into the harmonic foliation in the lower right through a series of different non-harmonic foliations in between. Within the same Whitehead class, there can be numerous non-harmonic foliations, but only one harmonic foliation.

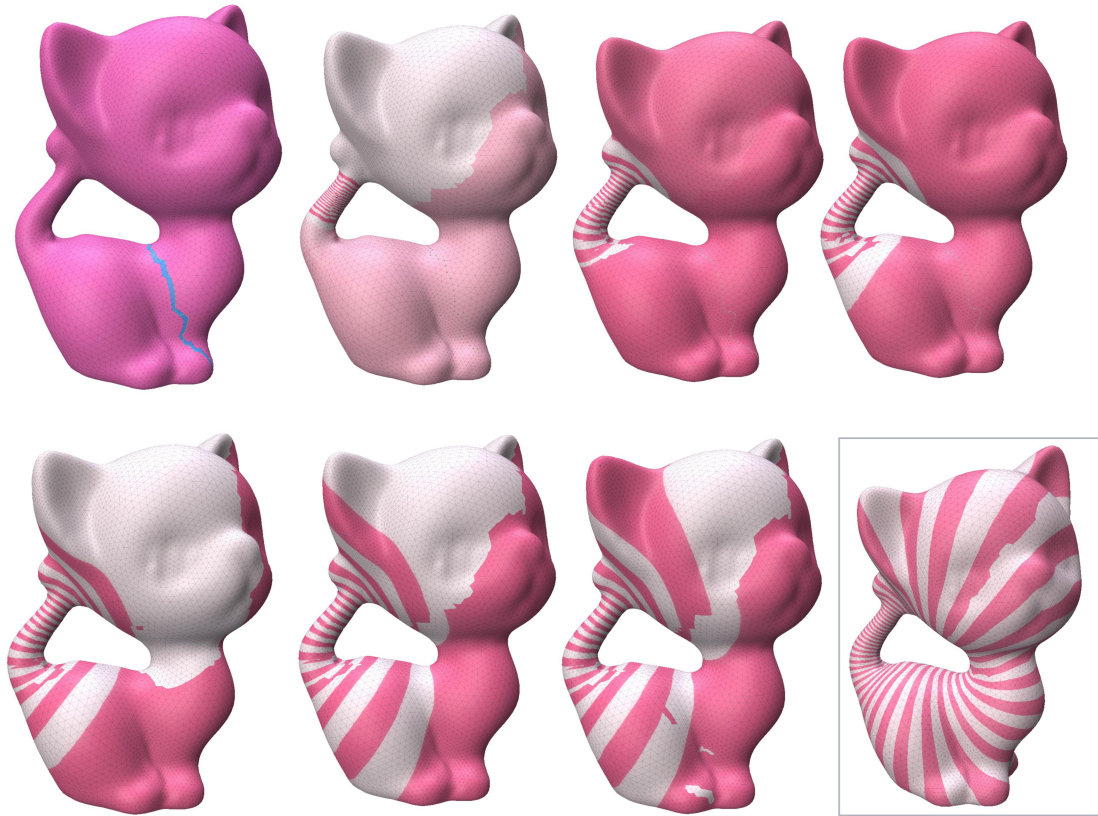
This optimization process does not exist in the field of mathematics and is difficult to derive from the perspective of abstract mathematical tools. However, from the perspective of algorithmic tools, it is quite natural to conceive of this method. This optimization process has inspired the idea proposed in this paper—using "algorithms" as tools to explore "global geometric structures"—which constitutes the core of computational discrete global geometric structures.

During the optimization of a non-harmonic foliation, support from Ribbon-Graph is required; otherwise, a harmonic result cannot be obtained. The optimization would get trapped in a local optimum, resulting in another foliation with additional poles that does not belong to the same Whitehead class.

The design of the algorithm requires a large number of technical tools, such as computer graphics rendering. For example, the stripes in the figure are rendered and presented using texture mapping technology in computer graphics.

The "harmonicity" of a harmonic foliation is defined with respect to the metric of the surface. For surfaces with the same topology but different metrics, the shape of the harmonic foliation on the surface will also differ.

A surface can have countless Whitehead equivalence classes. If the initial foliation does not belong to a specific Whitehead class, the resulting harmonic foliation will belong to another Whitehead class.



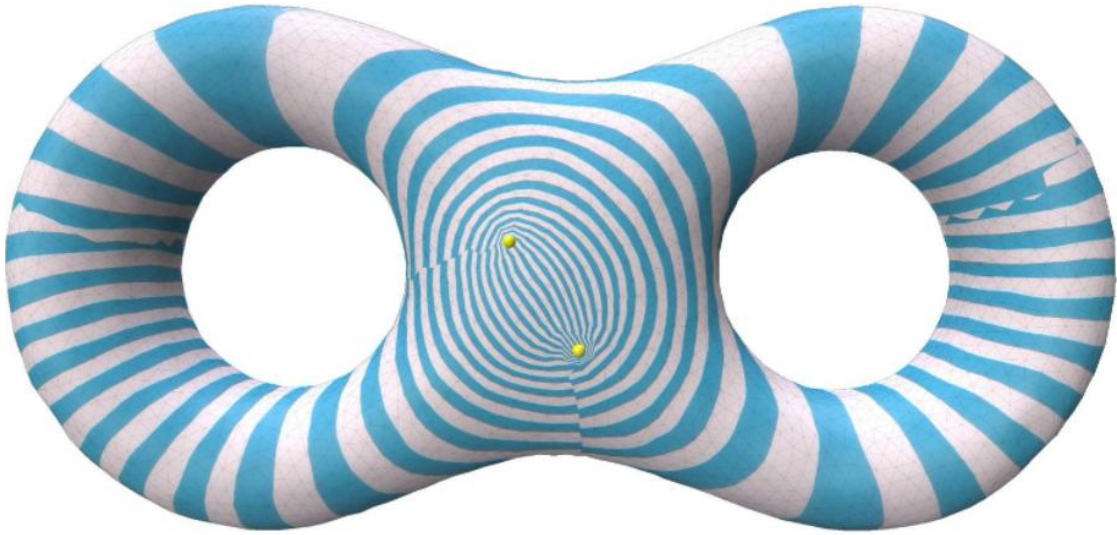
(Figure 11: A non-harmonic foliation to a harmonic one, generated by Geometric.)

(5) Harmonic Foliation with Poles

Harmonic foliations can also have poles, where the degree of the poles can be 1 (as shown in the figure below) or 2, etc. For a surface with genus g that contains poles, the harmonic foliation must satisfy the following equation:

$$\{\text{Number of poles}\} \times \{\text{Degree of poles}\} + \{\text{Number of zeros}\} \times \{\text{Degree of zeros}\} = (4g - 4)$$

A pole can be regarded as an infinitely small hole cut out of the surface. During the optimization process of a foliation from non-harmonic to harmonic, the positions of the poles remain unchanged, while the positions of the zeros keep moving [until they reach their optimal positions.](#)



(Figure 12: A harmonic foliation with two poles of degree one, generated by Geometric.)

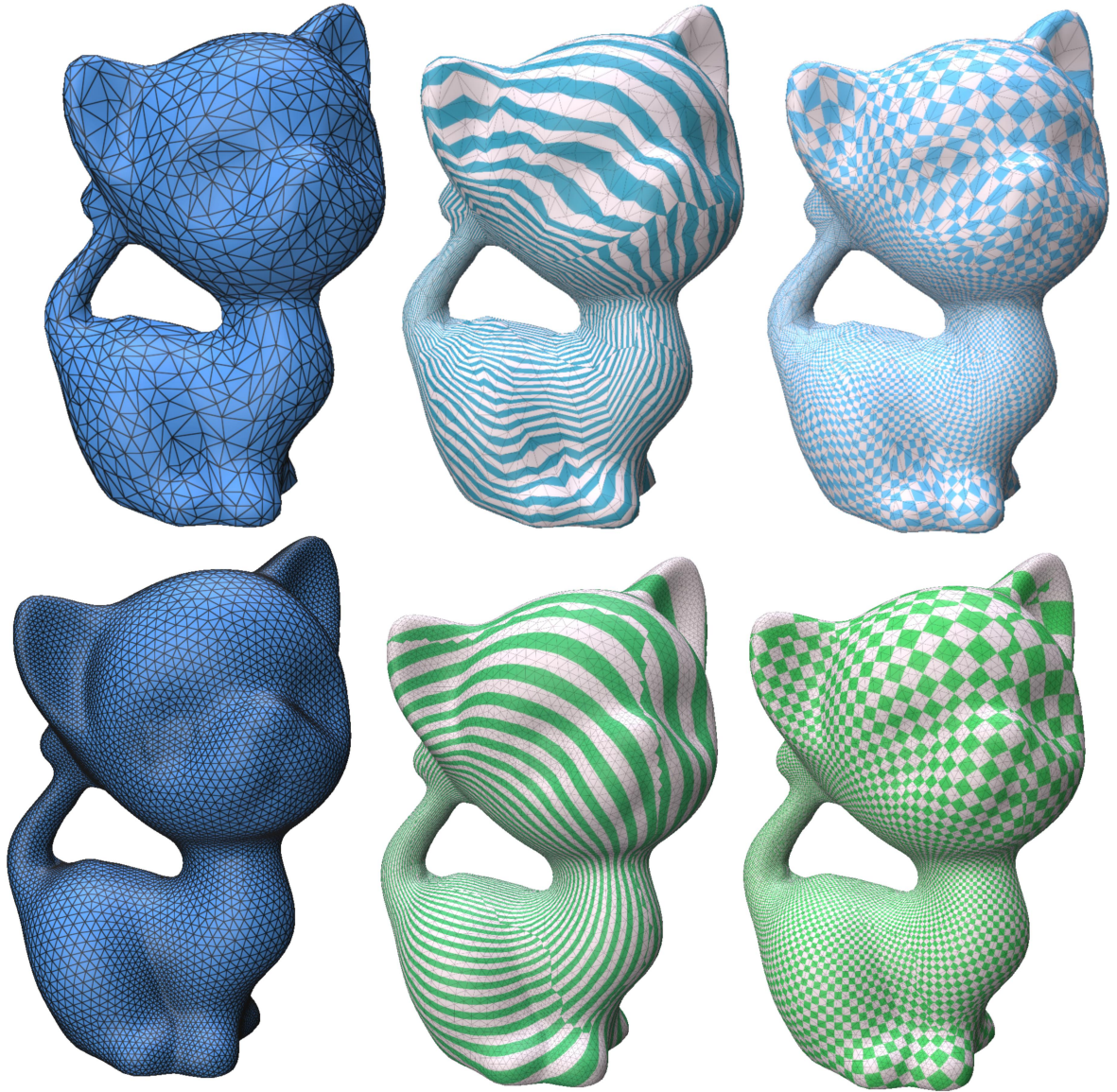


(Figure 13: A harmonic foliation with three poles of degree 2, generated by Geometric.)

(6) Holomorphic Quadratic Differentials

A holomorphic quadratic differential can be regarded as being composed of two mutually perpendicular harmonic foliations, as shown in the figure. The algorithm for this process first generates a set of horizontal harmonic foliations, and then generates the corresponding vertical harmonic foliations.

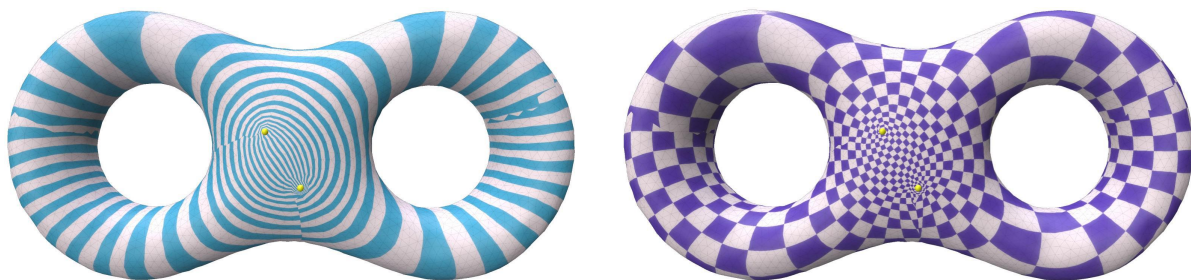
The poles and zeros of a holomorphic quadratic differential are consistent with those of the harmonic foliations; specifically, the zeros and poles of the two harmonic foliations (horizontal and vertical) are located at the same positions.



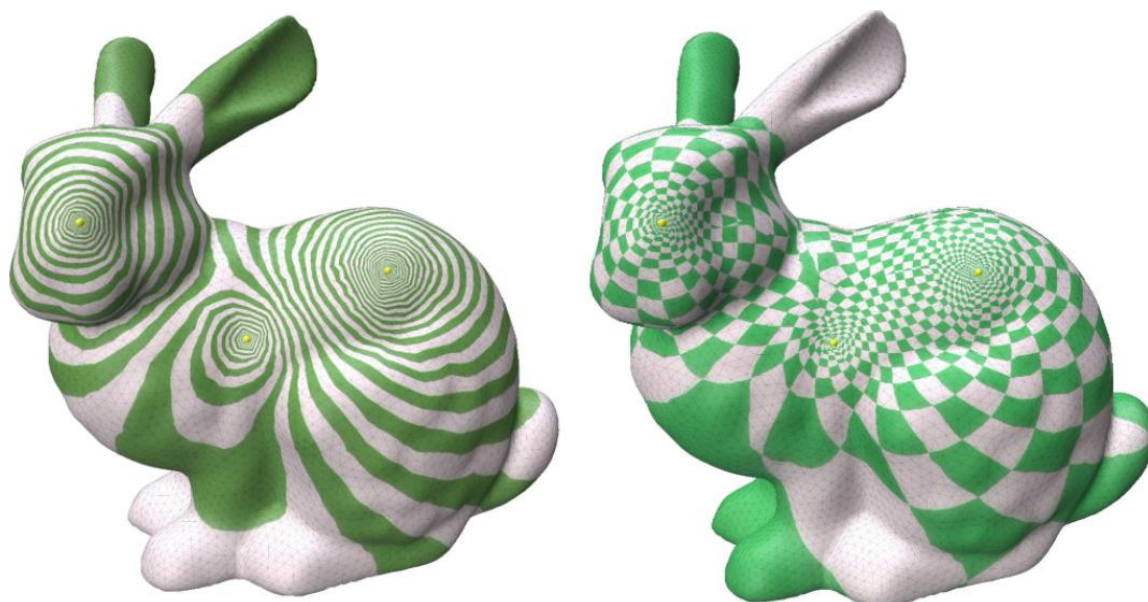
(Figure 14: A harmonic foliation and its holomorphic quadratic differential.)

(7) Meromorphic Quadratic Differentials

A holomorphic quadratic differential has only zeros, not poles. If a harmonic foliation has poles, the corresponding differential is a meromorphic quadratic differential.



(Figure 15: A harmonic foliation and its meromorphic quadratic differential, generated by Geometric.)



(Figure 16: A harmonic foliation and its meromorphic quadratic differential.)



(Figure 17: A harmonic foliation and its meromorphic quadratic differential.)

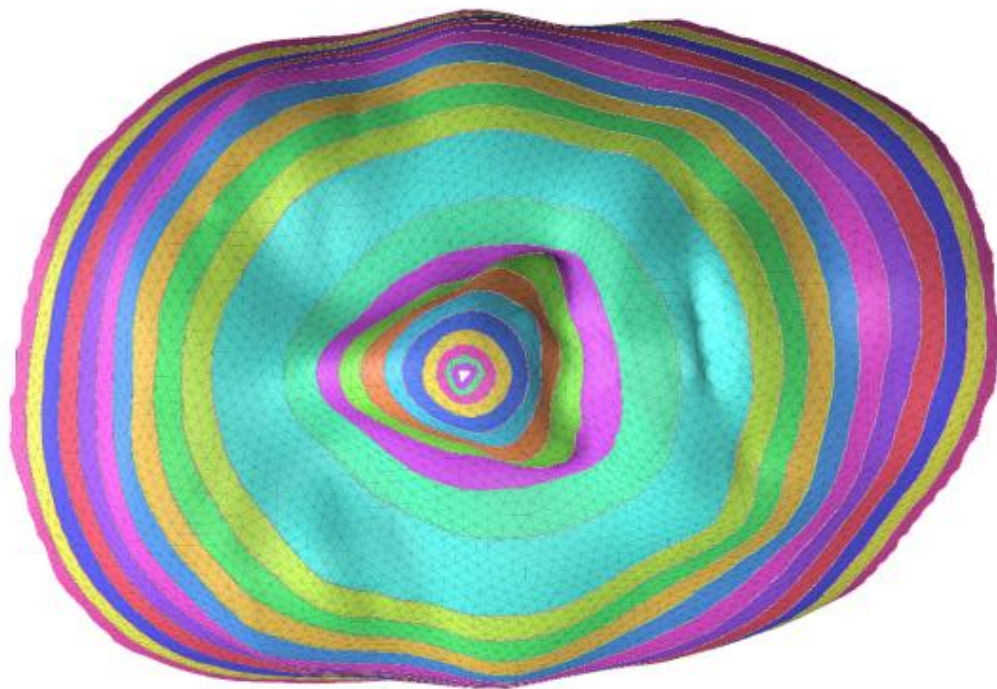
(8) Harmonic foliations on a surface with boundaries

The foliations and holomorphic quadratic differentials discussed earlier are all defined on closed surfaces; however, foliations can also exist on surfaces with boundaries. A surface with boundaries differs from a surface with infinitesimal holes (punctures). Previously mentioned harmonic foliations with poles have infinitesimal holes (punctures) at the pole locations on the surface, whereas the holes of a surface with boundaries are not infinitesimal—they have a certain perimeter. Therefore, it is impossible to treat the boundaries as poles, and consequently, it is not feasible to obtain corresponding meromorphic quadratic differentials.

As a result, the two cases (surfaces with poles and surfaces with boundaries) differ not only in mathematical concepts but also in the process of algorithm design. Although the general algorithm framework remains similar, detailed analyses need to be re-conducted. When expressed using mathematical formulas, both harmonic foliations with poles and those on surfaces with boundaries can be represented as harmonic functions in local coordinate systems; thus, it is difficult to capture their global properties from this local perspective. From an algorithmic perspective, however, since direct replication of existing algorithms is not possible and new algorithm designs must be considered, this provides a different viewpoint to explore the essence of foliations and gradually discover the characteristics of their global geometry. The process of designing these detailed algorithms and conducting repeated experiments has gradually inspired the authors of this paper to establish a research approach for the computability of global geometric structures.

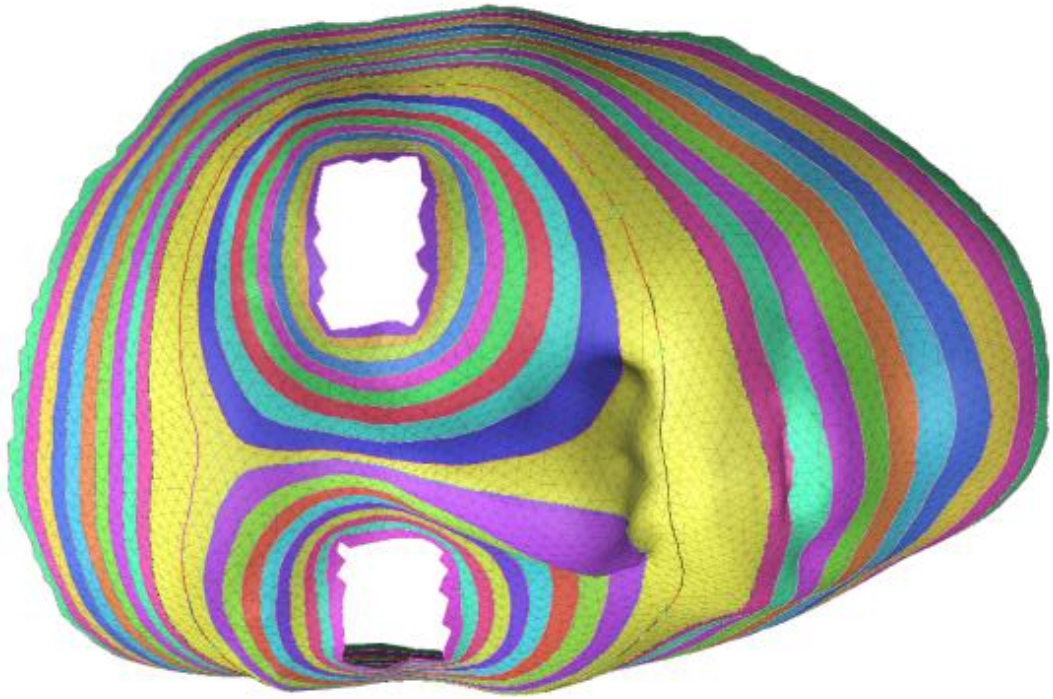
In the case of surfaces with boundaries, the annular regions along the boundaries can be treated as non-harmonic foliations for optimization, ultimately yielding harmonic foliations. Alternatively, a line can be drawn between two boundaries to serve as the initial foliation.

It is also possible to sew two identical surfaces with boundaries along their boundaries to form a complete closed surface (without boundaries). The resulting foliation will be consistent across both original surfaces. The sewn surface is a surface of higher genus; however, since the two original surfaces are identical, it may visually appear as a single surface in three-dimensional space. During algorithm design, computations can be performed on the sewn closed surface—only by reclassifying the original boundary annuli as non-boundary annuli in the new surface will the results remain

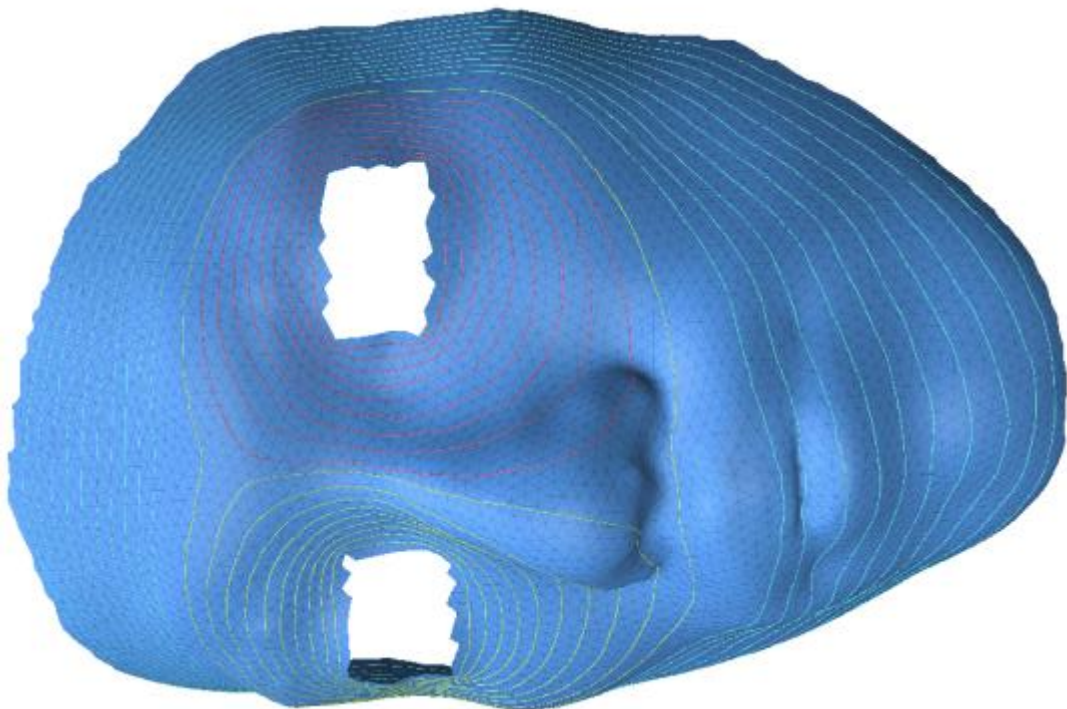


consistent.

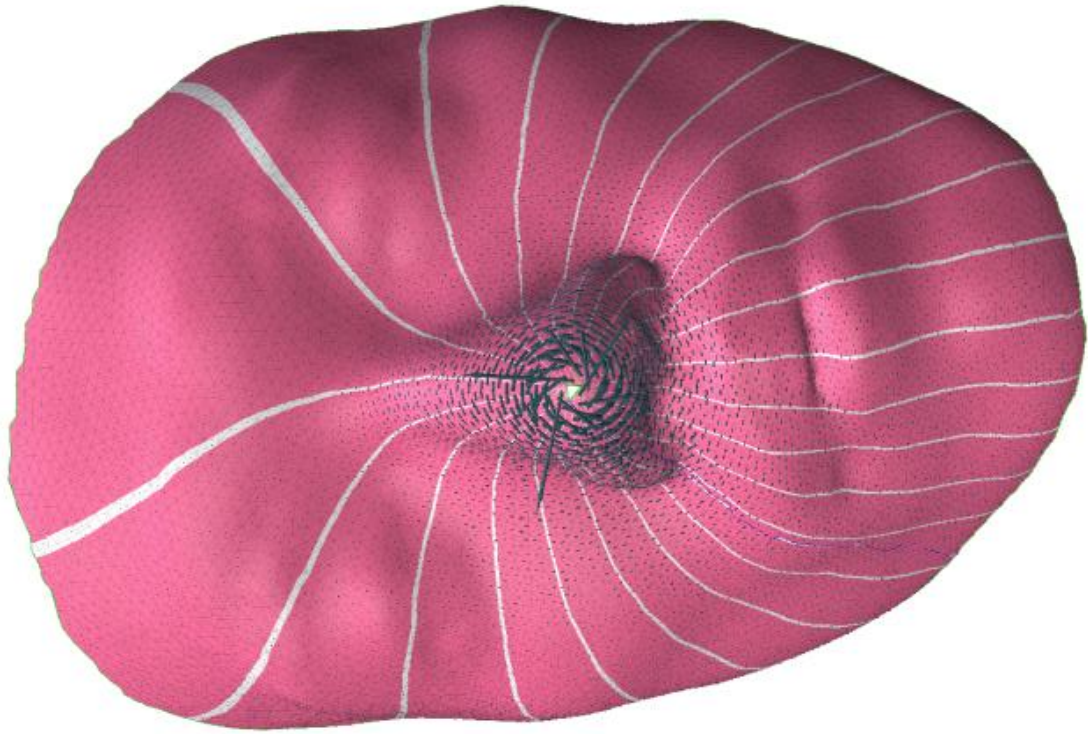
(Figure 19: A harmonic foliation on a surface with boundaries, generated by Geometric.)



(Figure 20: A harmonic foliation on a surface with boundaries, generated by Geometric.)



(Figure 21: A harmonic foliation on a surface with boundaries, generated by Geometric.)



(Figure 22: A harmonic foliation on a surface with boundaries, generated by Geometric.)

(9) Super Structured Quad Mesh

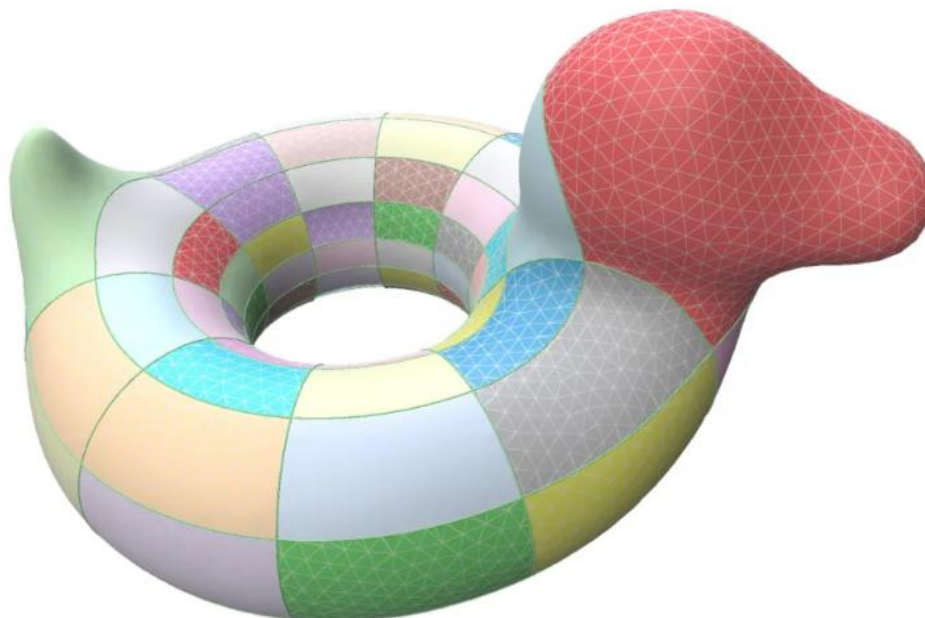
Examples of super structured quad meshes on genus-1 meshes are shown below. A surface with genus 1 has no zeros, so super structured quad meshes are relatively easy to obtain.

However, moving from genus 1 to higher genera, or when poles are present, entirely new theories need to be introduced.

A key point in quad mesh generation algorithms is that the quadrilaterals can be squares or rectangles. It is generally assumed that quadrilaterals need to be squares, leading to a research approach that employs conformal theories and tools—since conformality preserves angles, it facilitates the realization of square tiling. Nevertheless, both squares and rectangles are defined based on the metric of the surface. Therefore, this paper proposes focusing on the tiling arrangement structure of quadrilaterals, which is independent of the metric. After addressing this overall arrangement structure, the quadrilaterals can then be optimized into squares. As shown in the figure below, for rectangular tiling, research can be further conducted on how to optimize rectangles into squares under the current metric while maintaining the unchanged arrangement structure.

For high-genus meshes in particular, since zeros are unavoidable, the aforementioned arrangement structure becomes more complex. It is difficult to simultaneously solve both the arrangement problem and the shape optimization problem in one go.

In this way, the problem— which originally requires considering both the overall arrangement and the realization of squares — is decomposed into two independent subproblems. This enables a divide-and-conquer approach, where different theories and technical tools can be applied to each subproblem separately. This also constitutes an important contribution of this paper.



(Figure 23: Square Tiling, generated by Geometric.)

4. Computational Discrete Global Geometric Structures

The research on super structured quad meshes is an example and subfield of computational discrete global geometric structures.

In 2022, Zhao Hui proposed the research direction, content, field, tools, framework, and methods of "computational discrete global geometric structures," and elaborated on these concepts through channels such as WeChat official accounts. The term "computational discrete global geometric structures" can be parsed into four core components: (Computable) - (Discrete) - (Global) - (Geometric Structures), with each component defined as follows:

When computer scientists compute global geometric structures, their primary approach is to discretize smooth surfaces into meshes and then design algorithms for computation on these meshes. This is the origin of the term "(Discrete)".

- (1) The component "(Global)" emphasizes the focus of this new research direction, distinguishing it from the more familiar field of "(Local) differential geometry".
- (2) "(Geometric Structures)" clarifies that the research focuses on geometric properties rather than topological properties.
- (3) "(Computable)" refers to "algorithms" rather than simple formula-based calculations.

Global geometric structures are fiber bundle-based geometric structures defined on various manifolds, representing a more refined structure than topology. The author defined the concept of "computational discrete global geometric structures" and proposed this research direction, which primarily includes three aspects:

- (1) Developing algorithms to capture the global properties of various global geometric structures;
- (2) Visualizing various global geometric structures using computer graphics technologies;
- (3) Applying the aforementioned algorithms to various industrial and technical fields.

In most cases, researchers can directly derive algorithm designs from abstract concepts of global geometric structures. However, the example of super structured quad meshes demonstrates that the "algorithmic perspective" can reveal insights that are difficult to perceive from an abstract theoretical standpoint. Importantly, this perspective is not developed out of thin air—it relies on existing global geometric structures to generate new ideas.

This example illustrates that the "computational discrete global geometric structures" proposed in this paper is not limited to finding a one-to-one "algorithm" for a single geometric structure to capture its global mathematical properties. Instead, it requires integrating multiple "global geometric structures" to develop a comprehensive algorithmic framework. The relationship between computational discrete global geometric structures and global geometric structures is not a simple, mechanical one-to-one correspondence—and this is precisely the significance of this new research direction.

In his 1994 classic paper *On Proof and Progress in Mathematics*, William P. Thurston transcended traditional formalized mathematics to conduct in-depth discussions on the nature of mathematical research, the diversity of mathematical understanding, and the socialization of mathematical knowledge. Thurston argued that mathematical understanding is not a single logical process but a multidimensional, individualized cognitive activity. He identified seven distinct ways to understand the mathematical concept

of "derivative":

- (1) Infinitesimals (early ideas in calculus);
- (2) Symbolic computation (algebraic rules);
- (3) Logical definition (ε - δ language);
- (4) Geometric intuition (slope of the tangent line);
- (5) Physical rate (instantaneous velocity);
- (6) Linear approximation (optimal local approximation);
- (7) Microscopic limit (limiting behavior under high magnification).

These ways of understanding are not mutually exclusive but complementary and interdependent. The work of mathematicians extends beyond constructing rigorous definitions—it also involves reconciling different perspectives to enable the effective application of mathematical concepts in fields such as physics, engineering, and computer science. As Thurston emphasized, one of the core challenges in mathematics education is guiding students to internalize these diverse ways of thinking, rather than merely focusing on rote memorization of formulas and proofs.

Different understandings of a concept can lead to the development of new mathematical structures. For instance, the concept of exterior differentiation can be built on the fourth understanding (geometric intuition), but it is difficult to develop this concept from perspectives such as the third (logical definition). In other words, the goal of exploring multiple understandings of a single mathematical concept is to transcend the concept itself and enable new developments in diverse directions.

The proposal of the "computational discrete global geometric structures" research direction stems from the authors' observations of practical algorithms and experiments, which gradually coalesced into a concrete framework. For example, through researching algorithms to capture harmonic foliations with poles, the authors realized that in addition to the seven understandings of mathematical concepts proposed by Thurston (familiar to mathematicians), computer science's "algorithms" provide an additional dimension for understanding mathematical concepts. The key insight is to move beyond formalized memorization and transform concepts into a process of understanding and practical application.

For instance, the "holomorphy" of holomorphic quadratic differentials must satisfy the definition of conformality, and can thus be analyzed from a conformal perspective. However, the conformal perspective is a local one, making it difficult to grasp global properties or design algorithms that capture global constraints. These reflections were part of the authors' process of developing the "computational discrete global geometric structures" research direction, gradually establishing a framework that focuses on "global geometric structures" and uses "algorithms" as the core tool.

The development of the "computational discrete global geometric structures" research direction did not follow a linear sequence—i.e., mastering differential geometry first, then global geometry, and finally studying algorithms. Instead, it began with "algorithms": in the process of researching specific algorithms (initially unaware that these algorithms related to global geometric structures), the authors gradually synthesized various phenomena into a coherent framework, ultimately forming the philosophy and approach of using "algorithms" as tools to study "global geometric structures".

In the early stages of researching foliations, the authors were unaware that foliations are global

geometric structures, nor had they established the concept of "global geometric structures" itself. This concept emerged from practical algorithmic research, not from rote memorization of definitions in mathematics textbooks. While studying algorithms related to foliations, the authors repeatedly encountered phenomena where computational results could not be obtained through local formulas or approximation methods. Centering on "algorithms," they gradually explored relevant mathematical concepts and theories, eventually realizing that a unified framework could be formed by adopting the "global geometric structure" perspective and using "algorithms" as tools—paving the way for new research fields.

A key point for most science and engineering scholars is that the concept of global geometry is rarely encountered in conventional education, making it difficult to understand. The critical and obvious differences between local and global geometric structures are as follows:

If the topology of a manifold changes (e.g., a hole is punctured in a surface while the geometric shape of other regions remains unchanged), the local geometric values of those unchanged regions (such as Gaussian curvature) will not be affected.

In contrast, global geometric structures such as harmonic foliations will change entirely. This is because global geometric structures cannot be defined at individual local points—they are defined on the entire manifold.

Although global geometric structures can be represented by local point values under a specific coordinate system, altering the manifold (even if the local shape of a point remains unchanged) can cause these local values to change due to modifications in distant regions of the manifold (a "ripple effect").

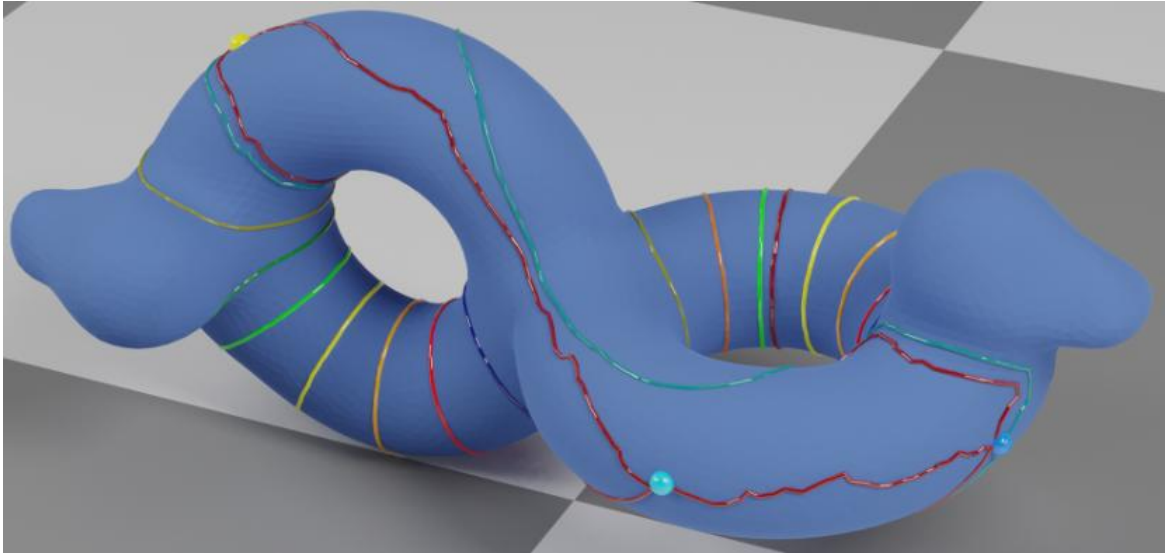
For example, global geometric structures such as differential 1-forms, vector fields, harmonic foliations, and holomorphic quadratic differentials will undergo complete changes if a small hole is punctured in the surface.

To date, the motivation for researching the computability of various global geometric structures has originated from applications in mesh processing within the field of computer graphics. However, with the proposal of the "computational discrete global geometric structures" research direction, its innovation points and analytical perspectives have been clarified. It is expected that the scope of this research will extend beyond computer graphics to find applications in interdisciplinary fields such as topological materials and biological genetics.

The authors' thinking on "computational discrete global geometric structures" evolved gradually, driven primarily by their research on algorithms for specific global geometric structures. Initially, they did not realize that their work should be framed within "global geometric structures"; instead, during the process of designing algorithms, they gradually became familiar with the mathematical properties of these global structures, mastered their global attributes, and eventually recognized that algorithms for many other geometric structures also fall under the umbrella of global geometric structures—thus warranting integration into a unified "global geometric structure" framework. The global geometric structure that most inspired the establishment of "computational discrete global geometric structures" is the first global geometric structure required for super structured quad mesh research: harmonic foliations.

Building on work by collaborators and mentors during his visiting research at the Harvard University School of Engineering—including Professor Steven J. Gortler, Dylan Thurston (son of Bill Thurston), and students supervised by Gortler—the author conducted in-depth algorithmic research on foliations of 2D surfaces. After years of effort, the author designed a convergent algorithm for generating harmonic

foliations.



(Figure 24: A harmonic foliation and Ribbon graph, generated by Geometric.)

Algorithm design serves as a tool for understanding global geometric structures. Through the algorithmic process of transforming non-harmonic foliations into harmonic ones, and using harmonic foliations as a concrete example, researchers can gradually grasp the global properties of a broader class of global geometric structures — along with their differences from local geometric structures — from an algorithmic perspective. Without concrete examples or hands-on experience in algorithm research, it is difficult to comprehend the core essence of underlying global geometric structures purely through abstract concepts.

The establishment of "computational discrete global geometric structures" is driven by its ability to solve problems that are intractable from other perspectives. For instance, differential 1-forms can be viewed as a subset of harmonic foliations, while harmonic foliations represent an extension of differential 1-forms. Algorithms for differential 1-forms can be designed using conformal geometry or exterior differentiation; however, these approaches rarely extend to algorithm design for foliations.

The core of computational discrete global geometric structures lies in algorithm research. Yet, due to the inherent properties of global geometric structures themselves, this field also fosters numerous groundbreaking innovations. This is primarily because current technology mostly relies on research into local mathematical structures, while global mathematical research has not yet been popularized in technical fields. Consequently, any algorithm related to global geometric structures is likely to drive progress and development across various technical domains.

The evolutionary path — from local differential geometry $\rightarrow$ topology $\rightarrow$ global geometry $\rightarrow$ global geometric structures $\rightarrow$ computational discrete global geometric structures — clearly demonstrates how abstract mathematics gradually advances toward industrial applications. This path aligns more closely with the inherent logic of scientific and technological development.

Global geometric structures are currently in a historical phase of evolution from abstract theory to computable implementation. Algorithmic research on computational discrete global geometric structures is

not a mere "calculation" of abstract global geometric structures. Instead, it first requires discretizing the manifold on which these structures depend, followed by algorithm design on the discrete manifold. The core objective is to preserve the global properties of global geometric structures (from smooth manifolds) during discretization, rather than merely achieving local approximation.

In the process of researching algorithms for various global geometric structures, new geometric objects and research tools are emerging—this is not limited to simple calculations based on existing mathematical theories. In fact, new tools and methods developed through the computability of certain global geometric structures have successfully solved many long-standing challenges in industrial technology. For example, the "index counting" technique proposed by Professor Steven J. Gortler (Harvard University) during his algorithmic research on the global geometric structure of differential 1-forms was later applied to the field of rigidity, addressing numerous technical problems in this area.

The research direction of "computational discrete global geometric structures" established by the author encompasses four key dimensions: theoretical research, algorithm design, code implementation, and industrial application. Its application scenarios extend beyond CAD-CAE-CAM industrial software to support research in fields such as topological materials in physics. Currently, many topological materials and quantum phenomena are, in essence, physical realizations of specific global geometric structures.

The academic community has generally recognized the profound role of abstract mathematics in physical experiments. Nevertheless, physicists face three practical dilemmas:

Difficulty in allocating extra time and energy to master cutting-edge geometric and topological theories;

Even if they master these theories, identifying the specific mathematical tools needed from a vast body of theoretical knowledge remains challenging;

Finally, they must explore concrete pathways to apply these theories to their own physical research.

All three steps require physicists to invest enormous time and effort, and cannot be resolved solely by relying on external resources.

To address this, the author proposes the following approach: first, design algorithms and implement computations for the global geometric structures involved in these physical fields; then, use these algorithmic results to guide the practical implementation of physical experiments. By treating algorithms as a bridge, a direct connection is established between abstract geometric theories and physical experiments—thus resolving the three aforementioned investment dilemmas faced by physicists.

Among the various global geometric structures studied in mathematics, some can be artificially constructed via computer algorithms, while others are naturally generated through physical processes in nature. For example, physical particles such as electrons and photons are global geometric structures constructed by nature, and nature itself can be regarded as a "computer."

However, a large number of global geometric structures remain confined to the realm of abstract theory: only abstract proofs of their existence exist, with neither corresponding physical entities nor algorithmic constructions realized. With technological advancement, it is expected that future innovations in physical materials or algorithmic breakthroughs will enable the construction of these structures. The approach proposed in this paper is as follows: for natural global geometric structures not yet "constructed by nature," algorithms can first be designed to study their various properties and potential construction methods. Based on the results of this algorithmic research, corresponding physical experiments can then be

designed—ultimately realizing the construction of entirely new global geometric structures using physical materials, and transforming more global geometric structures from abstract concepts into tangible realities.

Just as the application of natural global geometric structures (e.g., electrons and photons) has driven the development of technological civilization over the past two centuries, artificially constructed physical materials based on global geometric structures will undoubtedly lead to even greater advancements in technological civilization in the future.

This constitutes the core contribution and ultimate goal of the author's proposal regarding the research direction, methods, fields, and tools of "computational discrete global geometric structures."

5. One of the Top Ten Engineering and Technical Problems of the China Association for Science and Technology

The research on super structured quad meshes holds two key values: first, it has scientific significance in integrating numerous geometric structures and developing algorithms for these structures; second, it has engineering and technical significance—as the core engine of the roadmap proposed by the author to address the first of the "Top Ten Engineering and Technical Problems" identified by the China Association for Science and Technology (CAST), it helps solve long-standing pain points in industrial software, such as isogeometric analysis, spline surface generation, hexahedral mesh generation, T-spline generation, surface intersection, and frame field computation.

Since 2018, CAST has continuously organized activities to collect and release major scientific and technological problems. For the 2025 initiative, problems were rigorously evaluated against four criteria—frontier nature, leadership, innovation, and strategic importance—through standardized review, assessment, and voting processes. Ultimately, 10 frontier scientific issues, 10 engineering and technical problems, and 10 industrial technology problems were selected, serving as a "guiding compass" for the sustained production of original and disruptive technological achievements. The main forum of the 27th CAST Annual Conference was held in Beijing on July 6, 2025, where the 2025 major scientific issues, engineering and technical problems, and industrial technology problems were officially released.

Among these, the first of CAST's Top Ten Engineering and Technical Problems is titled "Integrated Algorithms and Theories for Design-Simulation-Manufacturing of Complex Models", with the following description:

"Breakthroughs and global leadership in China's basic industrial software depend on original innovation and guidance from mathematical theories, requiring systematic, forward-looking, and comprehensive research planning. China's CAD-CAE-CAM software industry now faces a rare opportunity to catch up with foreign industrial software—this stems from several intractable pain points in traditional geometric representation and simulation within mainstream international industrial software. The geometric kernels of international CAD software took shape in the 1970s and 1980s; over time, entire CAD systems have become extremely complex and unwieldy, making it impossible to replace their underlying data structures and kernels. As a result, these pain points remain unsolvable. At this historical juncture—where global design and R&D software is poised for generational upgrading—solving these pain points, developing next-generation integrated CAD/CAE geometric kernels, and leveraging the rapid development of China's smart manufacturing will make it fully possible to achieve a 'corner overtaking' of foreign software."

This engineering and technical problem encompasses the following key aspects:

- (1) The problem explicitly identifies algorithms and theories (rather than mere engineering implementation) as its core focus.
- (2) Technological breakthroughs in industrial software must rely on the innovative application of mathematical theories.
- (3) The engineering field requires a forward-looking perspective to formulate comprehensive overall planning.
- (4) Mainstream international CAD-CAE-CAM software currently faces specific technical problems that are difficult to resolve.

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- (5) It points out that the root cause of these problems lies in the underlying data structures of the software's geometric kernels.
 - (6) It sets the goal of overcoming these specific technical problems to develop next-generation geometric kernels, thereby creating new industrial software that outperforms current market leaders.

Among CAST's Top Ten Engineering and Technical Problems, this problem has a distinct characteristic: it is highly dependent on the innovation and guidance of mathematical theories. However, the scope of mathematical theories is extremely broad—it includes classical content such as calculus (founded by Newton and Leibniz centuries ago), high school-level geometry, and computational geometry, as well as cutting-edge fields like geometric topology (pioneered by Fields Medal laureates such as Shing-Tung Yau and William Thurston). Notably, CAST's description of this technical problem does not specify the exact mathematical theories required. Therefore, clarifying and exploring this aspect has become the primary research direction for solving the problem. This paper specifically proposes and recommends the exact mathematical theories needed to address the problem, aiming to move beyond vague discussions about "relying on mathematics" and advance relevant research into a substantive phase that is actionable, learnable, promotable, researchable, and implementable.

The research and processing object of CAD-CAE-CAM industrial software is complex 3D models, which inherently possess geometric forms. Thus, the core mathematical theories required to solve this engineering and technical problem in the field must be geometry-related. However, there are numerous branches of geometry-related mathematical theories. This paper clearly argues: the key to solving this problem lies in geometric theories related to "global geometric structures", rather than other geometric theories such as computational geometry or local differential geometry.

This does not negate the importance of other geometric theories in CAD-CAE-CAM industrial software. In fact, theories like computational geometry have long played a critical role in such software; after decades of development, their applications have become highly mature and have indeed made irreplaceable contributions to industry progress. The problem, however, is that the remaining engineering and technical pain points in the current CAD-CAE-CAM industrial software field are precisely those that existing theories (such as computational geometry) cannot overcome. Based on this, this paper explicitly proposes that CAST's engineering and technical problem targets unresolved core bottlenecks, and its breakthrough must rely on geometric topology theories related to "global geometric structures".

Brief Roadmap and Solution Proposed in This Paper (Based on Super Structured Quad Mesh Research)

- (1) It proposes that this engineering and technical problem requires cutting-edge geometric topology theories related to global geometric structures—rather than theories already maturely applied in CAD-CAE-CAM industrial software (e.g., computational geometry, local differential geometry).
- (2) It points out that the "geometric kernels" (geometric engines) in current industrial software are mainly based on the theories and technologies of computational geometry, while "mesh engines" primarily rely on local differential geometry.
- (3) It argues that the pain points caused by "geometric kernels" and "mesh engines" in mainstream international industrial software (e.g., issues in isogeometric analysis, spline surface generation, hexahedral mesh generation, T-splines, and finite element computation) cannot be solved by incremental improvements to existing "geometric kernels"; instead, they require a next-generation

"global geometric mesh engine".

- (4) It specifically proposes the concept of super structured quad meshes (with controllable overall quadrilateral arrangement structures) based on the application of these cutting-edge geometric topology theories, and suggests using super structured quad meshes as the core to solve the aforementioned pain points in CAST' s engineering and technical problem.
- (5) It designs a next-generation CAD-CAE-CAM industrial software architecture centered on super structured quad meshes, featuring a balanced focus on the "global geometric mesh engine," "geometric kernel," and "mesh engine"—to achieve the "corner overtaking" of domestic software over foreign competitors.
- (6) It proposes accumulating experience for systematic and forward-looking comprehensive research planning through initiatives such as the open-source meshDGP code, the Joint Research Center for Super Structured Quad Meshes, the "Mesh Roundtable" series of conferences, study sessions, and academic exchange activities on "computational discrete global geometric structures."
- (7) It takes solving this CAST engineering and technical problem as a driver to promote interdisciplinary integration and development across multiple domains—including teaching, research, application, popularization, art, visualization, and exchange—in fields such as geometric topology theory and application, mesh algorithm design, industrial software development, and mechanical analysis.

This roadmap and solution do not prioritize software product development; instead, it uses CAST' s technical problem as a motivation to drive the integration and development of interdisciplinary efforts in education, research, science popularization, programming, coding, visualization, and exchange across fields such as geometric topology theory, applied mathematics, computer graphics, mesh technology, and mechanical analysis.

The solution and roadmap proposed in this paper are not only relevant to future software but also hold significant value for CAD-CAE-CAM industrial software currently under development. They help avoid wasting substantial human, material, and financial resources on R&D for technologies that are either theoretically proven infeasible or not resolved.

Mathematical Theories in Current Mainstream CAD-CAE-CAM Software

Mainstream international CAD-CAE-CAM software development already relies heavily on mathematics, including theories such as computational geometry and numerical computation. Computational geometry primarily relies on mathematical principles such as high school-level plane geometry and undergraduate-level 3D Euclidean geometry; it achieves robust computational solutions to geometric problems through algorithm design. Since its core mathematical principles mostly fall within the knowledge system (from high school to undergraduate level) mastered by engineering experts, research in this field focuses more on the algorithm design aspect of computer science—integrating considerations such as computation speed, performance optimization, and memory usage to build efficient and reliable algorithm systems. In CAD-CAE-CAM industrial software technology, computational geometry is one of the core supporting technologies, with applications spanning the entire workflow from geometric modeling and analysis simulation to manufacturing.

Over the past few decades, many key technologies in CAD-CAE-CAM industrial software have relied on numerical computation for curves and surfaces. Examples include surface subdivision, surface simplification, parametric flattening, surface deformation, surface trimming, surface segmentation, and

surface reconstruction. For partial code implementations and effect demonstrations of these technologies, refer to the open-source mesh processing software meshDGP (as shown in Figures 4, 5, and 6).

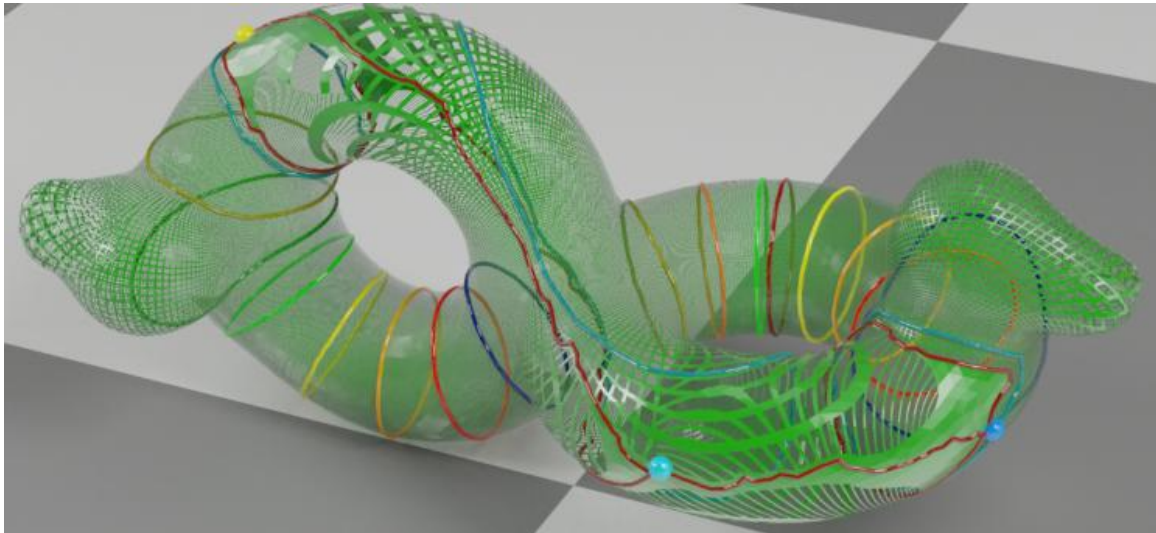
This research is mainly based on local differential geometry theory. As a core component of classical differential geometry, local differential geometry uses calculus to study the properties of 3D surfaces and 2D curves in Euclidean space. It focuses on the local characteristics of curves and surfaces and does not establish connections with the large-scale global properties in topology.

Since the theoretical system of local differential geometry had already matured by the 1950s, its diffusion from the mathematical field to the engineering field has been relatively thorough after nearly a century of dissemination. The engineering community has developed a solid understanding and mastery of concepts and theories related to local differential geometry; thus, research in this field focuses more on designing high-quality numerical algorithms to achieve efficient computation.

The shift from the Euclidean geometry theories (at the high school and undergraduate levels) underlying computational geometry to the calculus-based geometric theories of local differential geometry represents a significant leap in the adoption of mathematical theories. This also reflects that the development of CAD-CAE-CAM industrial software is a process of gradually incorporating deeper mathematical theories—especially geometric theories.

Currently, algorithms, code, and technologies based on local differential geometry theory already play a core role in CAD-CAE-CAM industrial software, serving as the foundation and indispensable components for commercial companies to develop such software. However, in practical use, the functionality of these software tools still fails to meet certain needs. The fundamental reason is that the theories underlying their technologies are inherently unable to address these needs. Therefore, the next phase of development for CAD-CAE-CAM industrial software urgently requires support from deeper mathematical theories—especially cutting-edge geometric topology theories. This is precisely the reason why CAST has identified this as an engineering and technical problem. Solving this problem is not merely an engineering and technical endeavor; its core lies in the introduction of deeper, more cutting-edge geometric topology theories.

The current research status of the engineering and technical problem "Integrated Algorithms and Theories for Design-Simulation-Manufacturing of Complex Models" is as follows: the specific mathematical theories required to overcome the problem have not yet been clarified—it is not in a phase where "the required theories are known, and only research and application are needed." The academic community is currently in an exploratory stage where this core premise remains "unclear." Therefore, the author's proposal that breaking through this problem requires geometric topology theories related to "global geometric structures" constitutes our primary contribution to solving this engineering and technical problem.



(Figure 25: A harmonic foliation and Ribbon graph, generated Geometric .)

5. Interdisciplinary Exchange of Geometric Topology Theories

Research on super structured quad meshes requires a large number of computer scientists who have mastered relevant frontier geometric topology theories, especially theories of global geometric structures. However, the number of scholars specializing in this field is relatively small. Therefore, we have organized a wide range of activities to promote the dissemination of these theories from the mathematics community to fields such as computer science, mechanics, and mechanical engineering. Conversely, these efforts also help geometric topology scholars understand how the abstract mathematical theories they study can enable potential technological applications through "algorithms."

Many mathematical masters and scientists have put forward an important assertion: "Basic mathematics is an engine for the development of high technology and can provide strong impetus for scientific and technological progress." This view has now been widely recognized by all sectors of society. What remains unclear, however, is how research on frontier mathematical theories can be transformed into a driving force for high-tech development. In the engineering field, while acknowledging the above assertion, the industry is more eager to explore this "transformation path" — that is, how frontier mathematical theories can specifically contribute to high-tech development.

A common current situation in the field of science and technology is as follows: The application of mathematical theories developed hundreds of years ago (such as calculus) has become highly mature, as these theories have long been widely mastered and skillfully applied by the engineering community. In contrast, new theories and tools developed by mathematicians in recent decades are still in the process of permeating and spreading to the field of engineering technology, and have not yet been popularized. This is a historical development stage that requires time to mature.

Even if experts in the engineering field clearly identify the required mathematical theories, it is by no means easy for them to additionally learn and master sufficient knowledge of relevant frontier mathematics within three to five years. The cognitive threshold for frontier mathematical theories is extremely high: A single new mathematical concept often requires 10 other new concepts as a foundation for understanding; each of these 10 concepts is in turn nested with 10 more basic concepts. Through such layered deduction, hundreds or even thousands of mathematical concepts that the engineering community has almost never been exposed to will soon emerge. For experts and scholars in the engineering field, they already need to invest a great deal of time and energy in research within their own fields, making it difficult to spare additional efforts to tackle these unfamiliar mathematical concepts. Only when there is a specific problem serving as a strong driving force can they be supported to complete this arduous learning process.

Scholars and experts in the field of engineering technology are usually familiar with the mathematical theories they encounter during their undergraduate and postgraduate studies, as well as their applications. Therefore, when facing specific problems, they often assume by default that the required mathematical theories are limited to this knowledge scope. Once the required mathematical theories go beyond this scope, engineering scholars face significant dilemmas in their practical work: They need to not only delve deeply into their own research fields but also invest an additional three to five years in learning and mastering more advanced mathematical knowledge. Under the realistic conditions of limited time and energy, this is obviously difficult to achieve.

Although the research plan and roadmap proposed in this paper—using super structured quad meshes to solve the engineering and technical problem of integrated algorithms and theories for complex models—

are centered on frontier geometric topology theories, the application of these theories is built on the foundation of our work in various aspects, including teaching, scientific research, science popularization, programming, coding, algorithms, software engineering, applied geometric topology, academic exchanges, scientific art, and visualization. The various innovations proposed to address this engineering and technical problem gradually emerged and developed during these work processes, through interactions and exchanges with experts in disciplines such as mechanics, mechanical engineering, mathematics, computer science, and physics.

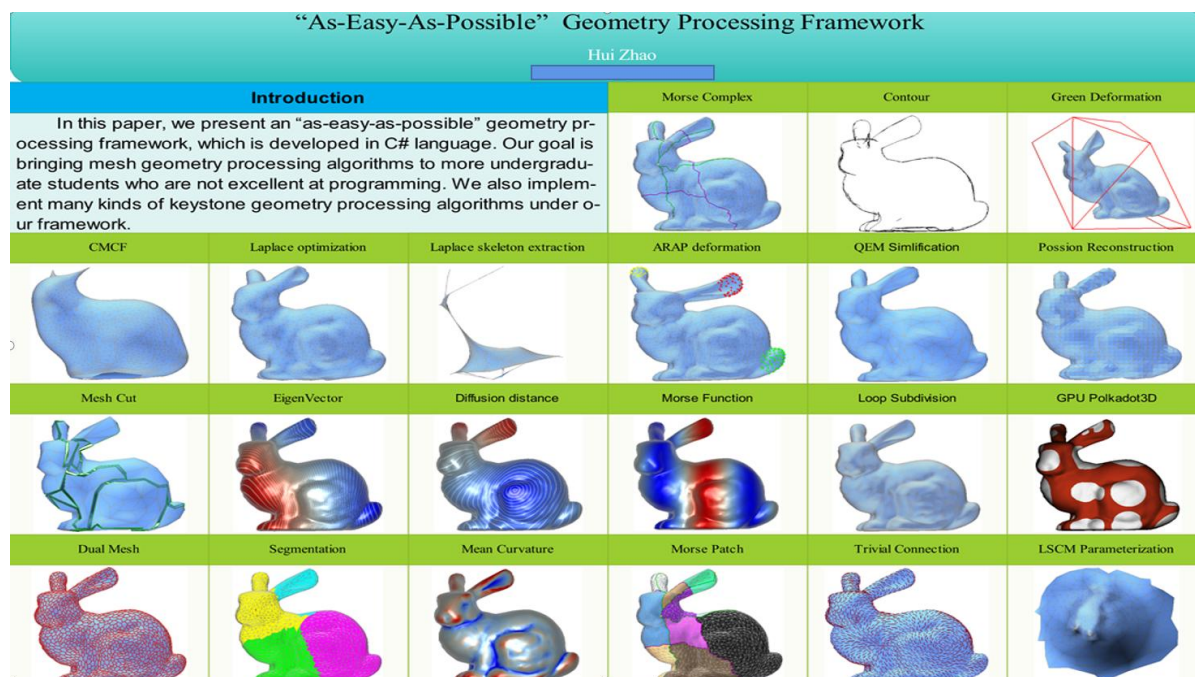
Therefore, we promote the dissemination of geometric topology theories to other industrial and technological fields through forms such as code, programs, visualization, art exhibitions, and seminars, and have organized a variety of academic and popular science exchange activities.

5.1 Open-Source Mesh Processing Code: meshDGP

Since 2007, we have independently developed the mesh processing software meshDGP. This software, consisting of hundreds of thousands of lines of code written in C#, integrates dozens to hundreds of various mesh processing algorithms, covering functions such as 3D model simplification, refinement, deformation, and parameterization, as shown in the figures below.

Designed for learning, research, and teaching purposes (rather than industrial applications), meshDGP adopts C# as its development language to lower the learning threshold. This not only allows beginners to get started quickly but also enables experts to focus more on algorithms themselves rather than programming techniques. The software is also integrated with interfaces for calling mainstream C++ model processing libraries (e.g., Libigl, Tetgen) and tools like MATLAB. Additionally, a supporting interactive interface and menu system have been developed to further enhance learning convenience. Meanwhile, its advanced software architecture and design patterns make it more suitable for students to learn algorithm principles and extend new functions.

According to rough statistics, approximately 40,000 to 50,000 science and engineering teachers, students, and engineers across China currently use this software for learning, research, and teaching activities. Over the past decade, it has been widely applied in teaching, research, and engineering practice across multiple disciplines, including computer graphics, mechanical analysis, mechanical engineering, differential geometry, virtual reality, biomaterials, CAD/CAE, finite element analysis, and industrial software.



(Figure 26: The demonstrations of many modling algorithms, generated by meshDGP.)

According to statistics from real-name registrations, tens of thousands of teachers, students, and engineers across interdisciplinary fields and industries are currently learning and using the meshDGP code. Their research directions include: Mechanical Engineering, Architectural Design, Numerical Analysis, Aeronautics and Astronautics Science and Technology, Applied Mathematics, GIS and BIM, Structural Engineering, Civil Engineering, Design and Media Engineering, Architecture and Civil Engineering,

Architecture, Computational Mathematics, Hydraulic and Hydroelectric Engineering, Intelligent Construction, Solar Physics, Computer Vision, Computer Science and Technology, Mesh Generation and Application, Electromagnetic Computation, Pure Mathematics, Numerical Solutions of Partial Differential Equations, Differential Geometry, Measurement and Control Technology, Conformal Geometry, Computer Graphics, Control Science and Engineering, Engineering Mechanics, Partial Differential Equations, Mesh Processing, Engineering Applications of Isogeometric Analysis, Stochastic Analysis, Optical Engineering, High-Performance Computing, Radio Physics, Intelligent Computing, Software Engineering, Automatic Control, Computational Fluid Dynamics, Finite Element Analysis, Signal and Information Processing, Geological Modeling, Physics, Aeronautical and Astronautical Engineering, Mechanics, Biology, High-Performance Computing for Multiphase Flow-Structure Interaction, Circuits and Systems, Medical Image Computing, Artificial Intelligence, Applied Statistics, Control Theory and Control Engineering, Structural Design, Automotive Engineering, Applied Physics, Innovative Mechanical Structural Design, Biomechanics, Medical Image Analysis, Quantitative Research on Architectural Form, Engineering Thermophysics, Nuclear Science and Engineering, Wind Turbine Aerodynamics, Mechanical Design and Theory, Materials Science, Geometry Processing, Extended Finite Element Method, Aircraft Design, Quasicrystals, Fractal Geometry, Graph Embedding, Structural Topology Optimization, Communications, Electronic Science and Technology, Fluid Mechanics, Electromagnetic Fields, Computer Security, and Department of Aerospace Engineering and Mechanics.

Five textbooks written by Professor Zhao Hui are associated with the open-source meshDGP code. By cross-referencing these five textbooks with the code, numerous teachers, students, and engineers have been able to quickly gain interdisciplinary entry into mesh processing-related algorithms in the shortest possible time. The textbooks are listed as follows:

- (1) Computer Graphics: An Introduction to 3D Model Processing Algorithms (C# Version), by Zhao Hui et al., Ocean Press, 2014.
- (2) Computer Graphics: OpenGL 3D Rendering (C# Version), by Zhao Hui et al., Ocean Press, 2016.
- (3) 3D Model Parameterization Algorithms: Theory and Practice (C# Version), by Zhao Hui et al., Publishing House of Electronics Industry, 2017.
- (4) 3D Model Deformation Algorithms: Theory and Practice (C# Version), by Zhao Hui et al., Publishing House of Electronics Industry, 2017.
- (5) Fundamentals and Examples of GLSL Rendering Programming (C# Version), by Zhao Hui et al., Publishing House of Electronics Industry, 2017.

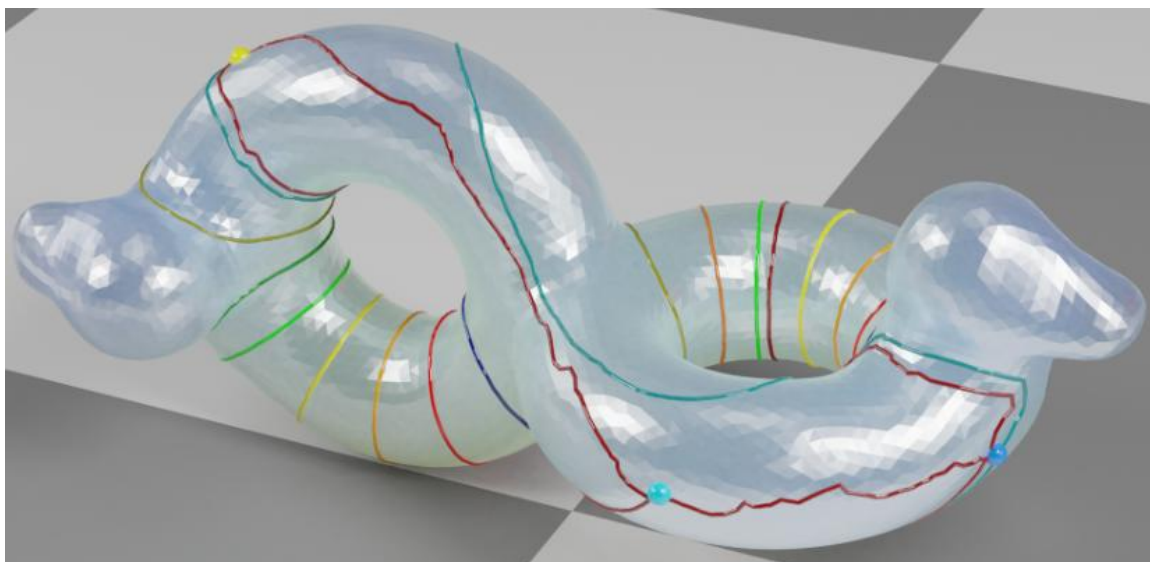


Based on real-name registration information, teachers, students, and engineers using meshDGP come from various institutions and enterprises, with some of the organizations listed below:

- (1) University of Shanghai for Science and Technology
- (2) Northwestern Polytechnical University
- (3) Yantai University
- (4) Xi'an University of Architecture and Technology
- (5) Kunming University of Science and Technology
- (6) Beijing University of Technology
- (7) Tongji University
- (8) University of Chinese Academy of Sciences
- (9) Xi'an Jiaotong University
- (10) Beijing Normal University
- (11) Zhejiang University
- (12) Tsinghua University
- (13) Xiangtan University
- (14) Zhongyuan University of Technology
- (15) Taiyuan University of Technology
- (16) Beihang University
- (17) Chongqing University
- (18) Hangzhou Normal University
- (19) Dalian University of Technology
- (20) Sichuan University
- (21) Jilin University
- (22) Jiangxi University of Science and Technology
- (23) Guangdong Pharmaceutical University
- (24) Henan Normal University

-
- (25) South-Central University for Nationalities
 - (26) National University of Defense Technology
 - (27) Beijing Union University
 - (28) Wuhan University of Technology
 - (29) South China Normal University
 - (30) Tianjin Polytechnic University
 - (31) Guangxi University
 - (32) Tianjin Chengjian University
 - (33) Changzhou Institute of Technology
 - (34) Central South University
 - (35) Zhengzhou University
 - (36) Nanjing Normal University
 - (37) Nanjing University
 - (38) Henan University
 - (39) China Institute of Water Resources and Hydropower Research
 - (40) Beijing Normal University
 - (41) Hunan University
 - (42) Beijing University of Posts and Telecommunications
 - (43) Xihua University
 - (44) University of South China
 - (45) Tianjin University of Technology and Education
 - (46) Harbin Institute of Technology
 - (47) Silicone Lake College
 - (48) Kashgar University
 - (49) Politecnico di Milano
 - (50) Iwate University
 - (51) Université catholique de Louvain
 - (52) Stony Brook University
 - (53) University of Liverpool (UK)
 - (54) The University of Hong Kong
 - (55) City University of Hong Kong
 - (56) Technion – Israel Institute of Technology
 - (57) École Polytechnique (Paris)
 - (58) China Academy of Engineering Physics
 - (59) National Supercomputing Center in Zhengzhou
 - (60) Hong Kong Cardiovascular Health Research Center
 - (61) Hefei Institutes of Physical Science, Chinese Academy of Sciences
 - (62) China Academy of Building Research
 - (63) Hong Kong Cardiovascular and Cerebrovascular Disease Research Center
 - (64) Jizhi Research Institute
 - (65) Sichuan Tianshang Youjia Network Technology Co., Ltd.
 - (66) Tengzhou Urban Construction Development Group
 - (67) Shenzhen Airport
 - (68) Liqing Automotive Technology (Shanghai) Co., Ltd.

-
- (69) Guangzhou Da An Gene Co., Ltd.
 - (70) Hubei Jiutongfang Microelectronics Co., Ltd.
 - (71) Shenzhen Conformal Technology Co., Ltd.
 - (72) Huawei Technologies Co., Ltd.
 - (73) Mihoyo Co., Ltd.
 - (74) Aibo Lingdong (Beijing) Technology Co., Ltd.
 - (75) Beijing Star Lan Technology Co., Ltd.
 - (76) Tencent Technology Co., Ltd.
 - (77) Jiangsu Jizhi Graphene Research Institute Co., Ltd.
 - (78) Yunnan Ciwei Biotechnology Co., Ltd.
 - (79) Zhengzhou Aviation Industry
 - (80) Shanghai Feijiegou Digital Technology Co., Ltd.
 - (81) Hubei Jiutongfang Electronics Co., Ltd.
 - (82) Beijing Kelaiming Technology Co., Ltd.
 - (83) Shanghai Huanshen Technology Co., Ltd.
 - (84) Zhejiang Zhihe Technology Co., Ltd.
 - (85) PIX MOVING
 - (86) Zhongwang Software Co., Ltd.
 - (87) Glodon Company Limited
 - (88) MediaTek Inc.
 - (89) Lenovo
 - (90) Guangxian Cloud (Hangzhou) Technology Co., Ltd.
 - (91) China National Offshore Oil Corporation
 - (92) ByteDance**



(Figure 27: A harmonic foliation and Ribbon graph, generated Geometric .)

5.2 Differential Geometry Teaching Based on meshDGP

The core mathematical theories of CAD-CAE-CAM industrial software cover multiple geometric branches, including computational geometry and differential geometry. While theories related to computational geometry have become relatively widespread, there remain significant gaps in the application and teaching of differential geometry in engineering fields. Across China, the number of non-mathematics programs offering differential geometry courses is limited, making it particularly important to lower the learning threshold of this discipline for non-mathematics majors. In fact, students from most majors often find differential geometry dull and obscure due to its complex formulas and abstract mathematical language, which creates considerable difficulties in understanding.

To address this issue, with the support of leaders from multiple universities, we have collaborated with teachers from institutions including the School of Science at Zhongyuan University of Technology, Henan University, Kunming University of Science and Technology, Beijing University of Posts and Telecommunications, Yantai University, Hohai University, and the University of Chinese Academy of Sciences to launch an innovative teaching reform seminar titled "Visualized Teaching of Differential Geometry Based on meshDGP." The seminar aims to enhance students' interest in learning differential geometry and help them gain a deeper understanding of the discipline's essence. Relying on the meshDGP platform, we present concepts and theorems in differential geometry through visualization, enabling students to quickly grasp abstract geometric concepts, theorems, and formulas.

2023 创新教改

meshDGP实验平台/《微分几何》

2023 第二届《meshDGP+微分几何》创新教改之技术入门和教改入门

会议简介

零基础开始学习meshDGP，技术交流普及

会议内容

1. 赵海川 北京师范大学 博士生 meshDGP技术入门详解一
2. 郭晋源 中山大学 本科生 meshDGP技术入门详解二
3. 冷雁 烟台大学 创新教改成果展示
4. 郝建兵 滕州城建开发集团 学习经验分享
5. 周瑞芳 中原工学院 创新教改经验分享
6. 自由交流、提问、专家答疑
7. 张家玲 昆明理工大学 创新教改会议预告

主持人 河南大学数学与统计学院 李娜

会议时间：2023.05.09 下午3:00

腾讯会议：401-4449-8649

密码：0509

微信群 腾讯会议

教改联系人：李娜

邮箱：mengyifei001@163.com



《MeshDGP+微分几何》 教学改革创新改革 第六届研讨交流会 之渤海方案



会议简介

《微分几何》课程理论程度高，推导繁琐，不容易透过概念理解几何性质。赵海川老师开发的 meshDGP 有 20 万 Csharp 代码，数千个函数，对微分几何的很多几何拓扑概念、定理、几何结构进行了可视化，有菜单方便交互操作，并且有五本配套教材，资料详细全面。从而可以方便学生快速理解相关抽象的几何概念、定理和公式。

一方面为学生进一步学习深入的几何拓扑知识打下基础；另一方面为计算机视觉、三维重建、CAD/CAE、几何建模、网格处理、虚拟现实、影视特效、医学成像、计算机图形处理和渲染、虚拟人、生物数据分析、机器人运动学、拓扑材料、力学分析、土木建筑、计算共形几何、可计算离散整体几何结构、等等高精尖工程技术应用打下扎实的微分几何理论基础。

会议内容

主持人 开场白

1. 曾珂 西安建筑科技大学 微分几何结合 meshDGP 在土木的应用
2. 郝建兵 滕州城建开发集团 meshDGP 带来的学习微分几何的兴趣滕州经验
3. 周瑞芳 中原工学院 meshDGP 微分几何教学在中原工学院的实践
4. 冷雁 烟台大学 meshDGP 微分几何教学在烟台大学的实践
5. 赵海川 北京师范大学 博士生 meshDGP 之 Calabi-Yau 图形编程实践

主持人 总结展望

主持人 烟台大学数学学院副院长 孙丰云

会议时间：2023.05.19 晚上 19:00

腾讯会议：726-507-884

教改联系人：冷雁

Email: lengyan3511@163.com



(Figure 28: Poster of the Differential Geometry Teaching Seminar)

Students often remain confused about the practical value of differential geometry when learning solely through concepts and theorems. However, with the meshDGP platform, they can directly observe the connection between differential geometry and real-world applications—such as using parameterization to (remap surface textures). This builds a bridge between mathematical language and computer language, helping students profoundly grasp the discipline's practical value. Not only does this attract students to engage in differential geometry-related research, but it also cultivates talent for the field. Modern differential geometry provides key mathematical tools for numerous engineering technologies, yet its high learning difficulty has led to a scarcity of practitioners. Given that industrial technology development cannot proceed without these tools, the meshDGP platform serves as a "stepping stone" to stimulate students' interest and promote wider application of differential geometry in engineering.

Currently, researchers in physics, chemistry, architecture, biology, computer science, mechanics, and other disciplines often rely on mathematical tools like Fourier transforms and neural networks to solve engineering problems. Nevertheless, many challenges remain unsolvable with these tools. If more scholars can learn differential geometry through algorithm platforms, it is expected to provide new ideas for addressing complex issues. For non-mathematics students, their relatively weak mathematical foundations make it difficult to master the knowledge system and framework of contemporary differential geometry through online self-study—unlike graduate students who have a solid grasp of fundamentals like Fourier transforms and neural networks. The meshDGP platform, however, helps them quickly understand this mathematical tool while developing learning and practical abilities at an early stage.

The meshDGP algorithm platform contains hundreds of thousands of lines of algorithm code, allowing students to directly access complete algorithm frameworks. Therefore, our teaching model not only imparts theoretical knowledge but also simultaneously trains students in coding skills. This "mutually reinforcing theory and algorithm" learning path broadens their employment prospects. Currently, cutting-edge engineering fields such as computer vision, 3D reconstruction, CAD/CAE, geometric modeling, mesh processing, virtual reality, film and television special effects, medical imaging, computer graphics processing and rendering, virtual humans, biological data analysis, robot kinematics, topological materials, mechanical analysis, civil engineering, computational conformal geometry, and computational discrete global geometric structures (see articles on the "computational discrete Global Geometric Structures" WeChat official account for details) have an urgent demand for differential geometry expertise. These fields require not only talent with theoretical knowledge but also practical professionals capable of translating theories into algorithms. The seminar's teaching model is committed to cultivating interdisciplinary interdisciplinary talent with both theoretical foundations and hands-on skills.



(Figure 29: Poster of the Differential Geometry Teaching Seminar)

Through the seminars on differential geometry teaching, we have successfully pioneered a new approach to differential geometry instruction in China and globally — particularly tailored for non-mathematics majors. Following these seminars, many universities have launched relevant courses incorporating differential geometry knowledge based on meshDGP. Teachers from various disciplines at these universities include not only those specializing in computer graphics but also interdisciplinary professionals in fields such as finite element computation, CAD/CAE, and differential geometry. Examples of such courses are listed below:

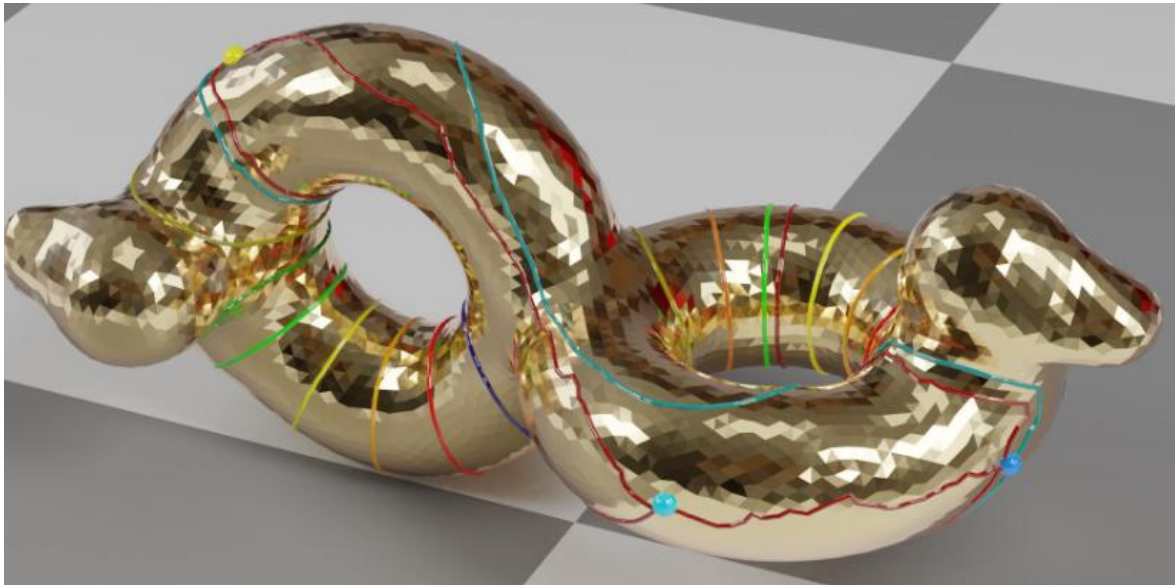
- (1) Kashgar University: Computer Graphics
- (2) Zhongyuan University of Technology: General Education Course - Introduction to Practical Differential Geometry
- (3) Beijing Yanqi Lake Institute of Applied Mathematics: computational discrete Global Geometric Structures (Fall 2022)
- (4) Yantai University: Differential Geometry
- (5) Xi'an University of Architecture and Technology: Engineering Modeling, Computer-Aided Design
- (6) School of Mechanical Engineering, Shandong University of Technology: Mechanical CAE and

Theoretical Courses

(7) Computer Graphics (Modernized Version)

Research on super structured quad meshes is inseparable from the widespread popularization of differential geometry and topology theories. To study a global geometric structure, one must accurately understand its mathematical concepts—and each such concept relies on 10 supporting concepts, which in turn each require 10 new concepts for comprehension. This quickly expands to hundreds of new mathematical concepts that most science and engineering teachers and students rarely encounter. Consequently, significant learning investment is required. For interdisciplinary teachers and students, it is essential to find methods suited to their backgrounds; they cannot simply replicate the learning approaches of mathematics majors, as they lack the corresponding learning environment, academic atmosphere, and incentive mechanisms available to mathematics departments.

The innovative seminars on differential geometry teaching we organized leverage tools such as meshDGP code and visualization to promote the dissemination of differential geometry across other disciplines. This, in turn, lays a knowledge foundation for the subsequent research and development of computational discrete global geometric structures. To date, positive results have been achieved: teachers and students at many domestic universities have become familiar with this approach, and it is expected that this new method of learning differential geometry and topology will gradually gain wider adoption in the future.



(Figure 30: A harmonic foliation and the Ribbon graph, generated by Geometric .)

5.3 National Traveling Art Exhibition of Computational Discrete Global Geometric Structures

The research direction of computational discrete Global Geometric Structures is not limited to studies in mathematics; it mainly targets research on applied geometry in computer science and other science and engineering fields. However, frontiers of differential geometric topology theories related to "global geometric structures" are rarely accessible to teachers and students of non-mathematics majors. Talents who understand and master these theories while being able to apply them in algorithm design are extremely scarce.

Although the geometric topology theories developed by mathematicians over the past decades have been gradually penetrating into engineering fields and applied in research practices, this interdisciplinary integration process is still in a stage of continuous development. Nevertheless, due to the high abstraction of these theories, teachers and students of non-mathematics majors often need a strong learning motivation to conduct in-depth exploration. The art form, however, can effectively stimulate the interest in learning and research among this group.

Based on the author's achievements in algorithm design in the application field of geometric topology theories, and with the help of rendering technology in computer graphics, these abstract geometric topology concepts and theories have been transformed into visual presentations. On this basis, in 2024, the author launched the [Global Touring Art Exhibition of computational discrete Global Geometric Structures](#), which has so far been held at more than ten universities and research institutions across the country, including:

- (1) [Zhongyuan University of Technology](#),
- (2) [Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences](#),
- (3) [North China Electric Power University](#),
- (4) [Henan University](#),
- (5) [Xihua University](#),
- (6) [Renmin University of China](#),
- (7) [South-Central Minzu University](#),
- (8) [Tianjin Chengjian University](#),
- (9) [Taiyuan University of Technology](#),
- (10) [High Magnetic Field Science Center, Hefei Institutes of Physical Science, Chinese Academy of Sciences](#),
- (11) [Jiangxi University of Science and Technology](#),
- (12) [Dali Li Zhengdao Science and Art Center](#).

The national touring art exhibition is not yet concluded, and dozens of more universities will host the exhibition in the follow-up.

The images exhibited cover geometric topology concepts such as Ribbon-graph, Square-Tiling, harmonic foliations, holomorphic quadratic differentials, meromorphic quadratic differentials, differential 1-forms, discrete Calabi flow, and normal flow.

During the exhibition, teachers and students from many universities feedback that this art exhibition has pioneered a new paradigm of "intrinsic geometric structure art" and serves as an innovative bridge connecting cutting-edge mathematical theories with public aesthetics. For the first time, the exhibition has systematically presented the intrinsic structures of surfaces that are difficult to perceive intuitively, transforming abstract theories such as holomorphic differentials and harmonic foliations into digital paintings with vivid colors, thus turning them from theoretical symbols into tangible visual entities.

Through this exhibition, the audience not only developed a strong sense of curiosity and desire for exploration but also truly felt the infinite charm generated by the interdisciplinary integration of mathematics and computer science. After visiting the exhibition, many teachers and students went to the library to look up relevant materials and learn about the mathematical concepts illustrated in the exhibition. Unconsciously, they developed an interest in these cutting-edge geometric topology theories.

Following the art exhibition, institutions including the [School of Science of Tianjin Chengjian University](#) and the [School of Science of Jiangxi University of Science and Technology](#) were still eager for more; they made full sets of reproductions of all over 50 artworks in the exhibition for collection and have launched long-term exhibitions on their campuses.

Compared with common exhibitions themed "Beauty of Mathematics" or "Geometric Art", this National Touring Art Exhibition has significant differences in both the core geometric and mathematical theories it presents and the external forms of presentation. Its core content lies in the "global geometric structures" generated through computer graphics technology and algorithms, and these geometric artworks are closely associated with applications in "chokepoint" fields such as industrial software, with the potential to solve relevant technical problems—which is exactly the unique value and innovative feature that previous lectures and exhibitions themed "Beauty of Mathematics" lack.

Although American mathematicians such as Thurston have popularized cutting-edge geometric theories and concepts through art forms, they were limited by the rendering technology available at that time and could hardly achieve the visual aesthetics supported by modern computer graphics. In addition, the fact that this exhibition is held as a national tour with the spontaneous participation of university teachers is an innovative attempt in itself.

Many teachers and students who came into contact with these differential geometric topology theories for the first time have summarized a large number of accurate insights and views on global geometric structures and computable research based on their own understanding, which has achieved the purpose of holding the exhibition.

Some teachers engaged in the research of these geometric topology theories also stated that the visual presentation is very helpful for their research work, as it can bring new perspectives and insights.

A number of schools have published graphic reports on the exhibition, including:

- (1) [Henan University](#),
- (2) [Renmin University of China](#),
- (3) [Tianjin Chengjian University](#),
- (4) [Taiyuan University of Technology](#),
- (5) [High Magnetic Field Science Center, Hefei Institutes of Physical Science, Chinese Academy of Sciences](#),
- (6) [Jiangxi University of Science and Technology](#).

Some schools have integrated the exhibition with [other mathematical activities](#), achieving even better results. For example:

[The 8th exchange and discussion activity of "RUC Mathematics Time I"](#) was held;

[The 8th Session of "RUC Mathematics Time I"](#) — Geometric Structures: Theory, Algorithms and Visualization.

Another example of integration with topological materials:

[The first science-art integrated art exhibition of Dali Li Zhengdao Science and Art Center opened at Taibao Home Dali Community](#).

“可计算离散整体几何结构”全国巡回艺术展

从 2023 年一月份开始，“可计算离散整体几何结构”实验室根据在图形学、应用微分几何拓扑、网格处理和生成等方面 30 年的教学、研究、工程工作，在国内首次提出通过各种艺术表现方法，例如图片、视频、雕塑、绘画等，以及用各种计算机图形学技术，例如虚拟现实、可视化等，来体现“当代微分几何拓扑”的数学理论和概念，尤其是“整体几何结构”的理论和概念。为了方便交流，我们把这种艺术称之为“当代几何拓扑”艺术流派。美国在上个世纪就有一些体现几何拓扑理论的计算机动画艺术，如 *Outside in*，*Not knot* 等，以及一些艺术雕塑。

“当代几何拓扑”艺术主要目的有两个，第一个是艺术层面，通过视觉信息来直观感受当代学者们开拓的美轮美奂的前沿的微分几何拓扑之美。第二个是工程科技层面，当前的卡脖子高科技需要对很多前沿的微分几何拓扑理论进行熟练掌握和深入应用，不仅仅是应用几百年前、一百年前的数学理论。而这些基础数学理论的掌握不是一蹴而就的，很多时候无法通过一个环节就能应用到工业技术上，需要一环扣一环，一个理论接一个理论的环环相扣的接力，最终走到工业技术产品层面。在中间阶段，没有工业产品激励的情况下，通过可视化、艺术展现带来的视觉美感可以对每一环理论的辛苦学习和掌握提供巨大的激励。

当前对基础科学数学理论和应用科学技术之间的关系有很多争议，主要原因就是有很多基础科学数学理论没有直接在第一个环节就被应用到工业技术上，所以很多朋友难以理解基础科学数学理论的重要性。“可计算离散整体几何结构”实验室提出艺术-科普-图示-算法-代码-理论一条龙思路，从而可以促进大家通过形象展示对某些基础理论有感性认识，进而理解基础理论如何通过一环一环的链条最终对工业技术发挥作用。

我们这次艺术展挑选了一些对调和一形式、调和叶状结构、全纯二次微分、微分一形式、法向量等若干“整体几何结构”进行可视化的、由我们研究或开发的计算机算法生成的图片。这些“整体几何结构”的理论，及其在 CAD/CAE 工业软件、拓扑材料、斯格明子物理材料、生物膜等交叉学科上应用的更多详细信息可以参阅微信订阅号“可计算离散整体几何结构”。



赵辉

2024 年 6 月 15 日



(Figure 31: Exhibition of computational discrete Global Geometric Structures at Zhongyuan University of Technology.)



(Figure 32: Exhibition of computational discrete Global Geometric Structures at the Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences.)



(Figure 33: Exhibition of computational discrete Global Geometric Structures at North China Electric Power University.)



(Figure 34: Exhibition of computational discrete Global Geometric Structures at Xihua University.)



(Figure 35: Exhibition of computational discrete Global Geometric Structures at Henan University.)



(Figure 36: Exhibition of computational discrete Global Geometric Structures at Renmin University of China.)



(Figure 37: Exhibition of computational discrete Global Geometric Structures at South-Central Minzu University.)



(Figure 38: Exhibition of computational discrete Global Geometric Structures at Tianjin Chengjian University.)



(Figure 39: Exhibition of computational discrete Global Geometric Structures at Taiyuan University of Technology.)



(Figure 40: Exhibition of computational discrete Global Geometric Structures at Jiangxi University of Science and Technology.)



(Figure 41: Exhibition of computational discrete Global Geometric Structures at the Hefei Institutes of Physical Science, Chinese Academy of Sciences.)



(Figure 42. Exhibition of computational discrete Global Geometric Structures at Dali Li Zhengdao Science and Art Center.)

Many scholars and experts have delivered lectures and reports on "Geometry + Art," "Mathematics + Art," or "Mathematics and Aesthetics." These reports typically share three characteristics:

- (1) For those who understand mathematical and geometric theories, beauty can be perceived "mentally, spiritually, and practically";
- (2) Beauty is visually perceived through the external "shapes" of certain geometric forms placed in 3D space;
- (3) Beauty is perceived through phenomena — for instance, natural processes like plant growth that conform to specific mathematical theorems, allowing audiences to appreciate mathematical beauty through observable occurrences.

However, the "Geometry + Art" presented in our national traveling art exhibition differs from the above three aspects in its expression of mathematical beauty. Its uniqueness lies in the fact that it does not showcase geometric "shapes" in 3D space, but rather "intrinsic" geometric "structures" on curved surfaces. Examples include harmonic foliations, Ribbon-Graphs, differential 1-forms, and Calabi Flow—these are not geometric shapes with tangible appearances and are generally invisible to the naked eye.

The innovation of this exhibition lies in using computer graphics technology to transform the beauty of intrinsic geometric structures into visual art. This artistic expression of geometric topology theories helps attract more teachers and students to learn and research these cutting-edge intrinsic geometric topology theories and their applications in industrial technology. The exhibition embodies the concept of "utilizing art as a powerful tool to promote the research and application of geometric topology theories." Prior to this, there have been some hand-drawn illustrations and computer-rendered images of intrinsic geometry internationally. However, systematically presenting intrinsic geometry through artistic expression in the form of an exhibition — especially a large-scale traveling exhibition — may be an unprecedented endeavor worldwide.

During the exhibition, the School of Science at Taiyuan University of Technology simultaneously held an online-offline press conference themed "Geometry and Art," inviting teachers from various majors at more than ten universities across China to give presentations. Participants shared insightful perspectives around topics such as geometry and art, science and art, and intrinsic geometry and art.

可计算离散整体几何结构全国巡回艺术展第九站：太原理工大学 新闻发布会

主持人	时间	内容	发言人	单位
嘉宾	16:00-16:10	领导讲话	武东升	山西发改委
	16:10-16:15	院领导发言	贺衍	太原理工大学数学学院
	16:15-16:30	太原理工大学全国巡回艺术展介绍	雷敏	太原理工大学数学学院
嘉宾	16:30-16:35	全国巡回艺术展第八站心得、体会、收获、经验	张东	天津城建大学理学院
嘉宾	16:35-16:40	全国巡回艺术展第七站心得、体会、收获、经验	朱剑林	中南民族大学计算机科学学院
嘉宾	16:40-16:45	全国巡回艺术展第四站心得、体会、收获、经验	韩喆	河南大学数学与统计学院
嘉宾	16:45-16:50	全国巡回艺术展第一站心得、体会、收获、经验	周瑞芳	中原工学院数学与信息科学学院
嘉宾	16:50-16:55	全国巡回艺术展第五站心得、体会、收获、经验	陈宏	西华大学机械学院
嘉宾	16:55-17:00	几何+艺术	孙志斌	中国科学院大学
嘉宾	17:00-17:05	几何+艺术	杨火根	江西理工大学理学院
嘉宾	17:05-17:10	科幻+艺术	陆贝珂	三体电视剧导演
嘉宾	17:10-17:15	数学+艺术	陈鹏	安徽师范大学
嘉宾	17:15-17:20	几何+艺术	刘绍辉	哈尔滨工业大学计算机科学与技术学院
嘉宾	17:20-17:25	内蕴整体几何结构 + 艺术	赵辉	可计算离散整体几何结构实验室
		现场师生交流+合影	师生	太原理工大学数学学院

5.4 Series of Mesh SquareTable Meetings

The algorithm research on computational discrete Global Geometric Structures focuses on meshes, investigating the corresponding global geometric structures on discrete meshes and smooth surfaces. Mesh research is not a purely abstract theoretical exploration; instead, it is a comprehensive study integrating the application of geometric topology theories, numerical computation, code implementation, visual rendering, and other dimensions. To achieve the goal of advancing the development of this field, with the support of teachers and deans from many universities across the country, the author established and launched the [Series of Mesh SquareTable Meetings](#).

It is worth noting that the reason this series of meetings is named "Square Table" rather than the common "Round Table" lies in the fact that square tiling is a core challenge in mesh partitioning. Consequently, the author proposed the concept of "super-structured quadrilateral meshes." "Super-structured quadrilateral meshes" refer to the research on dividing quadrilateral grids on surfaces using cutting-edge differential geometric topology theories such as various intrinsic global geometric structures. The shape of quadrilaterals highly aligns with the imagery of "square," so the name "Square Table" more accurately reflects the core of the research.

Up to now, eight sessions of the Mesh Square Table Meeting series have been held, with each session focusing on a different theme related to mesh research — such as code programming, industrial development, visual rendering, differential geometry teaching, optimization problems, solvers, interdisciplinary integration with computer graphics, paper writing styles, and mesh art. Each seminar attracts participants including teachers, doctoral students from different majors at various universities across the country, and engineers from enterprises. This series of meetings is still ongoing, and more mesh-related discussion topics will be covered in the future.

- (1) [The 1st Mesh Square Table Meeting](#): Code Programming Level
Hosted by the [School of Science, Tianjin Chengjian University](#)
Opening Speech: [Dean Jia Guozhi](#)
Organizer: Prof. Zhang Dong
- (2) [The 2nd Mesh Square Table Meeting](#): CAD/CAE Small Software Development Level
Hosted by the [School of Physics and Electronic Information, Anhui Normal University](#)
Opening Speech: [Dean Zhang Aiqing](#)
Organizer: Prof. [Chen Peng](#)
- (3) [The 3rd Mesh Square Table Meeting](#): Visual Rendering Level
Hosted by the [School of Science, Jiangxi University of Science and Technology](#)
Opening Speech: [Dean Xu Zhonghui](#)
Organizer: Prof. Yang Huogen
- (4) [The 4th Mesh Square Table Meeting](#): Code-Oriented Visual Learning and Teaching of Differential Geometry
Hosted by the [School of Mathematics and Statistics, Henan University](#)
Opening Speech: [Dean Tang Hengcai](#)
Organizer: Prof. [Han Zhe](#)

-
- (5) [The 5th Mesh Square Table Meeting](#): Optimization Problems, Linear Systems, and Solvers
Hosted by the [School of Mathematics, Statistics and Mechanics, Beijing University of Technology](#)
Opening Speech: Dean Huang Qiumei
Organizer: Prof. Zhuge Changjing
- (6) [The 6th Mesh Square Table Meeting](#): Interdisciplinary Integration of Computer Graphics + Other Disciplines
Hosted by the [School of Mathematics and Information Science, Zhongyuan University of Technology](#)
Opening Speech: Dean Yan Zhenya
Organizer: Prof. Zhou Ruifang
- (7) [The 7th Mesh Square Table Meeting](#): Evaluation of Paper Writing Styles in the Field of Mesh + Computable Global Geometric Structures
Hosted by the [Dean of the School of Science, Changzhou Institute of Technology](#)
Opening Speech: [Dean Chen Rongjun](#)
Organizer: Prof. Liu Yongcai
- (8) [The 8th Mesh Square Table Meeting](#): Mesh + Global Geometric Structures + Art
Hosted by the [High Magnetic Field Science Center, Hefei Institutes of Physical Science, Chinese Academy of Sciences](#)
Opening Speech: [Director Du Haifeng](#)
Organizer: [Director Du Haifeng](#)

The online live broadcast of each Mesh Square Table Meeting attracts 3,000 to 6,000 viewers online. This wide participation has effectively promoted the popularization of mesh research and laid a foundation for the algorithm research on "global geometric structures."

At present, the Mesh Square Table Meeting series is mainly held online. To further deepen research and exchanges in the mesh field (especially in the direction of super-structured quadrilateral meshes), Teacher Han from the School of Mathematics and Statistics, Henan University has applied for a conference project at the Tianyuan International Center for Mathematical Exchange, and Professor Yang Huogen from the School of Science, Jiangxi University of Science and Technology has also applied for a conference project at the Xiangshan Science and Technology Conference Center. Both conferences are planned to be held offline, which will create more direct and in-depth communication opportunities for participating experts and scholars, and facilitate the collision and advancement of research.

The purpose of the computational discrete Global Geometric Structures Laboratory in establishing the Series of Mesh Square Table Meetings is to address the second step of the assertion that "abstract mathematics and basic sciences are the foundation and engine of scientific and technological development," namely: "How to unleash the enormous potential contained in abstract basic mathematics, especially cutting-edge geometric topology theories, into technology, and promote the development of industrial technology."

Currently, the academic community, engineering sector, business community, and all sectors of society have reached a consensus on the first part of the assertion and recognized its importance. However, what everyone is eagerly looking forward to is the second step. Only through the second step can the first

assertion be truly interpreted, transforming it from a viewpoint into reality.

This is not a conceptual but a practical purpose. Abstract mathematics can be pursued with just paper, pens, and blackboards, but industrial technology requires "experimental tools." For example, the renowned physicist Samuel C. C. Ting needs to master every detail of the experiment, such as the voltage of each instrument; physical laboratories are so expensive that ordinary people cannot afford them.

In contrast, the research on "computational discrete Global Geometric Structures" on meshes proposed by us only requires a computer, which is affordable for tens of millions of science and engineering professionals. Our experimental technology is "programming." Nevertheless, even such experiments require mastering a large number of detailed experimental programming code details; otherwise, it would be mere armchair theorizing. One of the purposes of the Series of Mesh Square Table Meetings is to solve various practical problems such as experimental details, rather than just conducting exchanges on algorithm theories.



(Figure 43: The 1st Mesh Square Table Meeting)



(Figure 44: The 2st Mesh Square Table Meeting)



(Figure 45: The 3st Mesh Square Table Meeting)



(Figure 46: The 4st Mesh Square Table Meeting)



(Figure 47: The 5st Mesh Square Table Meeting)



(Figure 48: The 6st Mesh Square Table Meeting)



(Figure 49: The 7st Mesh Square Table Meeting)



(Figure 50: The 8st Mesh Square Table Meeting)

5.5 Organizing meshDGP Study Sessions

The research platform for computational discrete Global Geometric Structures is built on the original mesh code platforms such as meshDGP and Geometric. Since the release of meshDGP, tens of thousands of teachers, students, and engineers across the country have learned and used it. To address the question of "how to quickly get started with meshDGP using spare fragmented time", we have successively held more than 80 real-name themed study sessions. Through these study sessions, over 80 teachers, students, and engineers from various universities across the country and with different research directions have efficiently mastered the key points for getting started in their spare time, laying a foundation for in-depth research. Some of the study topics and participants are as follows.

题目	学习人
MeshDGP里的Morse函数	潘安 兰州
MeshDGP里的特征向量计算展示	瞿悦呈 北航计算机学院博士生
MeshDGP里面的QEM简化算法展示和代码流程	冯骊晓 老师 重庆科技大学计算机学院
MeshDGP里的向量场展示和代码流程【1/3】	赵晴 中原工学院数学学院 本科生
MeshDGP里的Calabi-Yau曲面编程	叶然然 本科生 中原工学院数学与信息科学学院数学与应用数学
MeshDGP里面的参数化效果图展示、代码概述：频谱，ABF算法篇章【1/2】	殷思麒 湖南大学机械学院 博士生
MeshDGP里的OpenGL展示和代码流程【1/2】	康亚卓 北京 中机认检
MeshDGP里的拉伸能量算法展示、代码概述【1/4】	丁弘晖 人大统计学博士生
MeshDGP里的OpenGL展示和代码流程II	潘安 兰州
MeshDGP里的GLSL效果展示和代码流程	祁桐 北京构力科技有限公司
四边形六面体网格生成之Polycube Shap Space	瞿悦呈 北航计算机学院博士生
基于Stretching Energy的变形实验展示	潘安 兰州
meshDGP剪影等操作的初步认识和后续学习目标	杜志楠 中国航空制造技术研究院
meshDGP里的GLSL渲染二刷	胡志林 博士 清华精密仪器系毕业
meshdgp里面的高斯曲率，平均曲率，欧拉示性数，亏格，点，边，面展示和代码流程	朱施恩 威斯康星-麦迪逊大学 UW madison 统计专业 研究生
线性代数之——MeshDGP里的OpenGL之矩阵变换展示和代码流程	孙克争 广东省科学院智能制造研究所
meshDGP软件架构和设计模式	康亚卓 潘安 彭贵军 黄锦龙 祁桐 vs 曾珂 西安建筑大学
meshDGP里如何实现样条曲面【1/2】	宋洋 博士 犹他大学 师从样条前辈教授
meshDGP里的骨骼动画和蒙皮展示与代码流程	肖智文 博士 清华航院
meshDGP里如何实现细分subdivision曲面【1/2】	杨火根 教授 江西理工大学
meshDGP里的贝塞尔曲线展示和代码流程【1/3】	银润龙 老师 忻州师范学院数学系
meshDGP里的简化光滑等操作和代码流程【1/3】	陆云桥 工程师 上海正雅齿科
meshDGP里的梯度，拉普拉斯与双调和操作和代码【1/2】（PPT代替）	刘亚雄 湖南
meshDGP里的三角形、四边形网格生成展示和代码流程【1/2】	王少阳 工程师 制造行业
meshDGP里的B-Spline曲线展示和代码流程【1/3】	黄锦龙 工程师 深圳 激光行业
meshDGP里的三维模型点边面的增加删除操作展示和代码流程【1/2】	陈鹏 老师 安徽师范大学物理与电子信息学院
meshDGP里的三维模型补洞、边界操作展示和代码流程【1/2】	王卫 中学老师 世青国际学校
meshDGP里的刚体转动 通过OpenGL【1/3】	姬振华 老师 郑州航院力学
MeshDGP里面的5,6,7度数操作展示和代码流程	李林 老师 太原科技大学机械工程学院
meshDGP里的matlab调用	刘宇辉 中南大学 老师 人工智能方向
meshDGP里的网格切割操作展示和代码流程	杨佩桥 北航电子信息工程 研究生
三维模型参数化扭曲度量及和谐参数化理论代码展示（无视频 使用PPT）	李鹏 北航
meshDGP里的补洞挖洞操作和代码流程	詹国峰 工程师 三维技术相关行业
MeshDGP里的ArcBall里面的（线性变换）原理和选择功能展示和代码流程	段育鹏 哈尔滨工程大学 博士生
meshDGP里的人机交互和UI	刘婧媛 东京大学五十岚 健夫Takeo Igarashi教授实验室博士后 香港科技大学visgraph lab博士
meshDGP里的线性代数50个应用之一	诸葛昌靖 北京工业大学数学学院 线性代数老师
MeshDGP里的三维模型特征线条抽取算法功能展示和代码流程	刘永才 老师 常州工学院理学院
meshDGP里的poission重建效果展示和对照代码	曾珂 教授 西安建筑大学
meshDGP里的法流操作展示和对照代码	古鹏举 博士 贵州大学
meshDGP里的poission重建操作展示和对照代码	王川川 工程师 宁波
meshDGP里的polycube变形	陈晨曦 博士 南洋理工大学
meshDGP里的GLSL渲染	孟楠 教授 香港大学医学院
meshDGP概览和研究生指导辅助工具	孙志斌 教授 博导 中国科学院大学 中国科学院国家空间科学中心
meshDGP里的三维模型分段算法展示	邵弘毅 博士 航发商发上海
meshDGP里的贴图和featureLine学习	伍世聪 工程师 北京+苏州

The Study Sessions are an innovative interdisciplinary learning and exchange model established by the computational discrete Global Geometric Structures Laboratory, with several prominent features:

- (1) Utilizing spare time;
- (2) Making use of fragmented time;
- (3) Enabling quick access to basics (quick getting-started);
- (4) No restrictions on academic discipline or major;
- (5) No requirements for prior background knowledge;
- (6) Learning with reference to code;
- (7) Visualized, menu-based, and interactive learning;
- (8) Participants who have already 入门 (mastered the basics) guiding new beginners;
- (9) No time or energy burden on either party—if either side feels burdened, the Study Session is deemed unsuccessful;
- (10) A completely different model from specialized in-school learning;
- (11) Particularly suitable for teachers and students with clear learning objectives;
- (12) Suitable for teachers and students with strong self-directed learning abilities;
- (13) No requirement for light workload in one's regular job—nearly 100 real-name registered teachers, students, and engineers who got started through the Study Sessions all did so in their spare time amid busy work;
- (14) Relatively suitable for teachers and students who are good at communication, as those who help with learning also do so in their spare time (on a voluntary basis), rather than being a teacher's regular job in school which involves guided instruction.

With its visualized interface, concise and clear-format code, convenient menu interaction, five supporting textbooks, and code-level application of a large number of mathematical theories, meshDGP is highly favored by a wide range of science and engineering teachers, students, and engineers across the country, and has especially gained recognition from many senior engineers. Judging from the enthusiastic participation of senior engineers from various industries in the Study Sessions, we have put forward the goal of "equipping 50 million science and engineering engineers with computational discrete Global Geometric Structures".

Currently, many cutting-edge computational science and technology are still in the stage of development and diffusion, which is a historical evolutionary process. Due to the limitations of the times and conditions back then, many senior engineers failed to access cutting-edge computing technologies in a timely manner. However, through the meshDGP Study Sessions, they can make up for this technological generation gap and achieve "looking back and catching up" in the course of technological development.

Over the past 50 years, many people among several generations in China have aspired to become scientists and engineers. However, with the changes in social development, this trend has weakened. In the future, Chinese teenagers may be more inclined to engage in fields such as finance. This is similar to the current situation in the United States where many engineering and technology talents come from other countries, or like the gradual reduction of the peasant population in China decades ago, with fewer people willing to engage in agriculture. Therefore, the core of our proposed goal of "equipping 50 million science and engineering engineers with computational discrete Global Geometric Structures" is to focus on the older generation of science and engineering engineers trained in China over the past 50 years, rather than only on current teenagers. When senior engineers master cutting-edge computing technologies, they will naturally form a demonstration and inheritance effect, laying a solid foundation for the growth and development of teenagers, and achieving a "natural and smooth" cultivation of technical talents.

At present, nearly 100 teachers, students, and engineers are conducting real-name online learning through the Study Sessions, accounting for one-fifth of all 500 real-name registered teachers and students, and 1% of the tens of thousands of teachers and students who use meshDGP in total. According to feedback from the teachers and students participating in the Study Sessions, basically, only three half-days of preparation in spare time are needed to get started. This proves that the mechanism of our Study Sessions is practical and feasible, conforming to actual conditions.

5.6 Joint Research Center for Super-Structured Quadrilateral Meshes

The research content of computational discrete Global Geometric Structures covers professional fields such as differential geometric topology theory, computer algorithm design, code implementation, and mechanical engineering. It is not a single algorithm; therefore, it requires interdisciplinary and cross-disciplinary collaboration, and tackles key challenges through an "organized scientific research" model.

Since the Laboratory of computational discrete Global Geometric Structures proposed the research direction of "super-structured quadrilateral meshes" to solve technical problems related to industrial software, it has attracted the attention of many professional teachers, scholars, and experts across the country, and relevant research has gradually unfolded. To advance "organized scientific research", the author has jointly established the "Joint Research Center for Super-Structured Quadrilateral Meshes" with multiple universities.

The joint mechanism of colleges from multiple universities fundamentally avoids the "five-only" tendencies (such as "only focusing on papers, only focusing on projects") that hinder scientific research development. The core value of the Joint Research Center lies in: taking "super-structured quadrilateral meshes" as the common research core, integrating the expert resources from multiple disciplines of various universities, and promoting the formation of a closed loop among multiple elements including "education, scientific research, science popularization, code, programming, programs, interfaces, interaction, visualization, art, geometric topology, mathematics, computation, algorithms, engineering, software, and academic exchanges". This realizes a positive cycle and iterative development of mutual promotion. Through the value demonstrated by the achievements, methods, and models themselves, it completely breaks the current "five-only" dilemma in the scientific research field, accelerates the transformation of the innovative achievements of human thinking contained in abstract mathematical concepts and theories into the field of science and technology, and ultimately contributes significantly to the development of high technology at home and abroad.

The specific objectives of the Joint Research Center include:

- (1) Taking the specific application of super-structured quadrilateral meshes in industrial software as an example, intuitively demonstrating the promoting role of "cutting-edge basic mathematics" in "high technology";
- (2) Demonstrating how to unleash the huge productivity contained in cutting-edge mathematical theories through specific application scenarios;
- (3) Establishing a communication channel for teachers and students from different schools through a specific platform;
- (4) Conducting cooperation in academic papers, code programming, industrial applications, and other aspects;
- (5) Organizing relevant academic and science popularization activities;
- (6) Studying the mathematical theories involved in super-structured quadrilateral meshes;

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- (7) Studying code implementation;
 - (8) Inviting experts in algorithms required for super-structured quadrilateral meshes to give exchange reports;
 - (9) Understanding the research ideas of computational discrete Global Geometric Structures through the case of super-structured quadrilateral meshes;
 - (10) Taking super-structured quadrilateral meshes as the core, laying a technical foundation for the next leapfrog development of domestic industrial software such as CAD and CAE.

At present, the School of Science of Jiangxi University of Science and Technology, the School of Science of Tianjin Chengjian University, and the School of Mathematics and Statistics of Henan University have successively passed the review of the college affairs meeting and obtained the signature confirmation of the dean, officially becoming the initiating units and core members of the Joint Research Center. Each of these three institutions has selected several teachers and graduate students to participate in the research work; up to now, the total number of teachers and students in the Joint Research Center has reached more than 20.

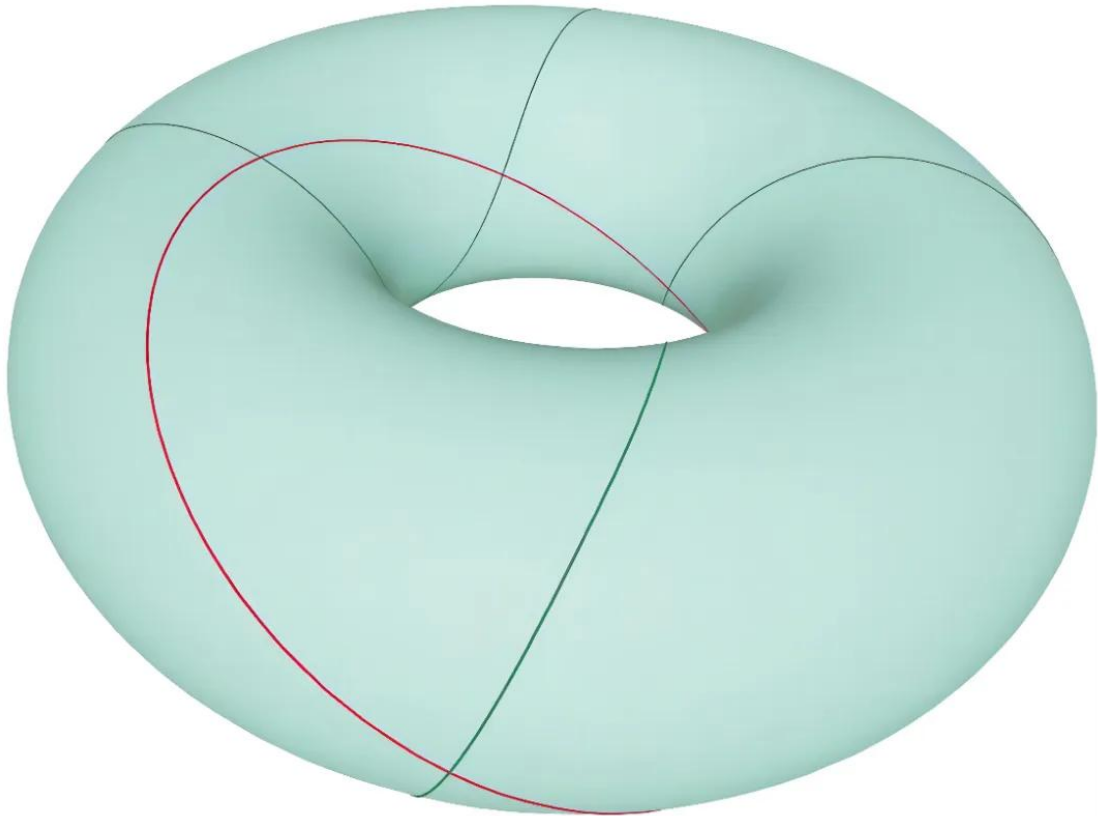
The research goal of computational discrete Global Geometric Structures is to release the productivity contained in the global geometric structure theory into various industrial technologies through algorithms. Compared with local mathematical structures, which have been relatively widely and maturely applied in industry, the productivity of global geometric structures is still in the embryonic stage and has broad prospects. This cannot be achieved overnight or by one person alone. Through large-scale organized scientific research, organizing teachers, students, engineers, scholars, and experts from various disciplines and professional backgrounds, it is expected to achieve mutual assistance, mutual exchange, and mutual improvement, thereby promoting the development of this new research field.

Super-structured quadrilateral meshes are a type of global geometric structure within computational discrete Global Geometric Structures. Due to their industrial application value, a platform such as the "Joint Research Center" was established first to provide further support for interested teachers and students. There are also various other global geometric structures, and in the future, organized scientific research may be carried out for them in a similar manner.

5.7 Super-Structured Quadrilateral Mesh Programming Competition

Computational discrete Global Geometric Structures is not merely a research on abstract mathematical theories, but a scientific and technological research that takes programming code debugging as an experimental tool. This means the research process requires in-depth and extensive support from code programming technologies and tools, especially the specialized programming capabilities in the mesh field—just as physics research cannot do without various physical equipment and experimental operations.

However, code programming, as a research tool, is often overlooked in this field: people tend to think it can be mastered without adequate preparation, and it also does not receive sufficient attention in the research process. Based on our practical experience, mesh-related code programming must be supported by adequate tool reserves, and researchers need to be proficient in various mesh programming technologies. For this reason, to cultivate this key technical capability, we have launched the "[Super-Structured Quadrilateral Mesh Programming Competition](#)" and set specific programming tasks. By completing these tasks, participants can systematically master most mesh-related programming technologies, laying a tool foundation for in-depth research.



(Figure 51: Schematic Diagram 1 of the Mesh Programming Competition Questions.)

The specific problem description is as follows:

Input a 3D model (closed) of a torus composed of triangles (which can be generated from mathematical formulas using the methods in the five textbooks);

First generate a quadrilateral torus, then generate a triangular torus;

Input an $N \times N$ grid on the torus (generated according to the torus formula);

The grid has N horizontal lines and N vertical lines (no need for the complex non-horizontal lines shown in the above figure; simple horizontal and vertical lines are used);

Cut the triangles on the torus into $N \times N$ small models along the grid;

Each small model is composed of triangles;

Render these small models in different colors;

Output these small models as files in OBJ or OFF format;

The diagram of the cutting process for the entire problem is shown in the duck diagram below.

Notes on the programming problem and its difficulties:

10. The programming difficulty lies in the fact that cutting the grid requires proficient mastery of the operations of the half-edge data structure for meshes (refer to the five textbooks for a large number of basic operations);

11. This programming task does not require algorithms or theories; it mainly tests the ability to write code;

12. A large number of "if-else" statements are needed to handle various cases of grid intersections;

13. Cutting along the horizontal and vertical lines of the grid requires passing through triangles, and each of these triangles will be cut into several smaller triangles;

14. Since all elements are generated by formulas, the intersection points between the grid and each triangle on the torus are already known during input;

15. Triangles that are not passed through do not need to be modified;

16. There are a large number of "if" cases that cannot be considered in the first place, and they need to be addressed while writing the program;

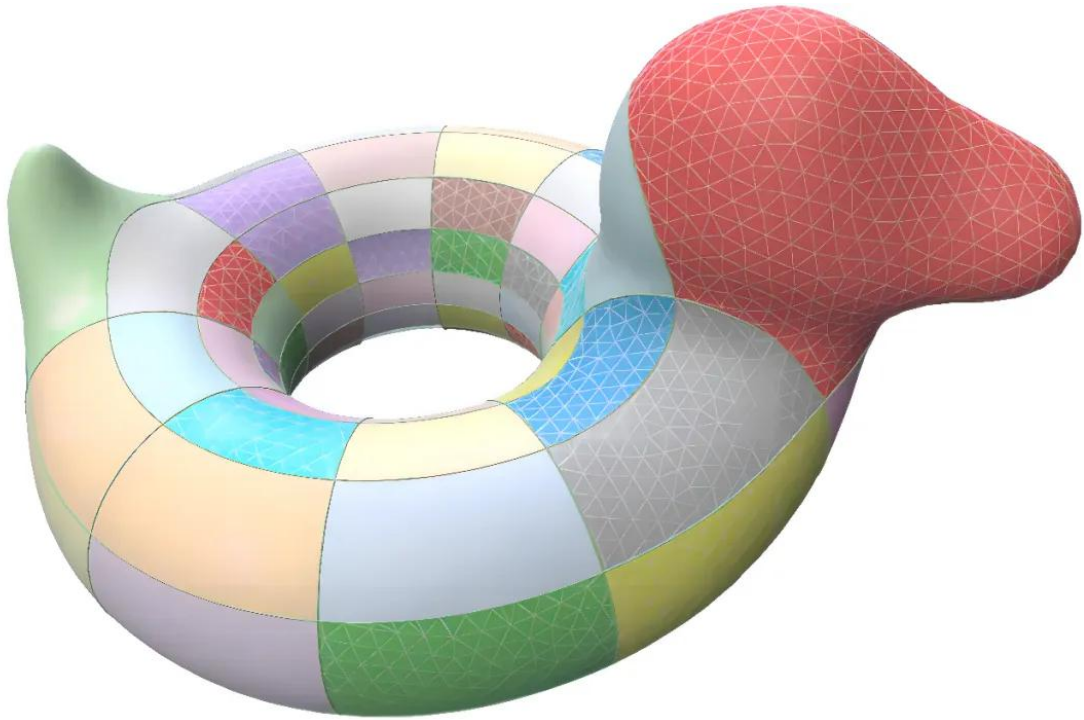
17. Numerical errors need to be handled through programming. For example, a horizontal line may be very close to the vertex of a triangle, so robust coding skills are required to determine how to cut through the triangle;

18. The key challenge is that the coding for the intersection of horizontal and vertical lines is extremely complex (it is estimated that dozens of "if-else" cases need to be considered), and the process requires great caution. Only after the cutting is successfully implemented will one feel a sense of relief (like a breeze);

19. Similar to the old oil peddler in the story, proficiency comes from practice;

20. Based on existing mesh programming experience, completing this section of code may require 3 to 4 weeks of immersive, full-time programming;

21. A large number of detailed UIs and rendering techniques are needed for visual assistance during the cutting process, which makes it easier to identify bugs visually (refer to the UI of meshDGP and the OpenGL and GLSL content in the five textbooks).



(Figure 52: Schematic Diagram 2 of the Mesh Programming Competition Questions.)

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